A GENERALIZATION OF BOMBIERI'S EFFECTIVE BIRATIONALITY THEOREM IN HIGH DIMENSIONS

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Abstract. We prove that on normal projective varieties, adjoint linear systems of nef and big Cartier divisors define birational maps with the constants of Fujita's conjecture, and so do pluricanonical systems when the canonical divisor is Cartier, nef, and big. In particular, we solve Questions 5.10 and 5.11 of Han's paper on Fujita's freeness conjecture, and we prove the birationality part of Catanese's conjecture on pluricanonical maps.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Fujita's conjecture [Fuj87] predicts that for a smooth projective variety X of dimension d and an ample Cartier divisor L on X, the divisor $K_X + (d+1)L$ is globally generated and the divisor $K_X + (d+2)L$ is very ample.

The global generation part of Fujita's conjecture was proved for $d = 2$ in [Rei88], for $d = 3$ in [EL93, Corollary 2*], for $d = 4$ in [Kaw97], and for $d = 5$ in [YZ20], see also [YZ17]. In any dimension, [AS95, Corollary 0.2] proved that $K_X + mL$ is globally generated for any integer $m \geq d(d+1)/2 + 1$, [Hei02, Theorem 1.4] proved global generation for any integer m at least a bound of order $d^{4/3}$, and [GL24, Theorem 1.1] proved global generation for any integer $m \geq d(\log\log d + 2.34)$ when $d \geq 2$. For several points, [AS95, Corollary 0.4] proved that $K_X + mL$ separates any r distinct closed points for any integer $m \geq (d^2 + 2rd - d + 2)/2$. See also [GL24, Theorem 1.3]. Bounds quadratic in d for the birationality of $|K_X + mL|$ were obtained in [EKL95, Corollary 3] and [DC14]. For $d = 3$, [GL24, Corollary 5.3] proved that $|K_X + 5L|$ defines a birational morphism onto the image. [Han26, Theorem 1.1] proved global generation for any integer $m \geq \lceil C_0 d \rceil$, where $C_0 = 1.77629988\ldots$. [OAI26] proved the global generation part of Fujita's conjecture in any dimension, see Theorem 1.3.

We now turn to pluricanonical systems. [HM06, Tak06, Tsu07] proved that for any positive integer d there exists a positive integer m_d depending only on d such that $|mK_X|$ defines a birational map onto the image for any integer $m \geq m_d$ and any smooth projective variety

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X of general type of dimension d. By [ETW23, Theorem 1.1(2)], any such integer m_d is at least $2^{2^{(d-2)/2}}$ when $d \geq 2$. Catanese proposed the following pluricanonical analogue of Fujita's conjecture.

Conjecture 1.1 ([Cat22, Conjecture 0.1 a) and b)]). *Let X be a minimal projective variety of general type of dimension d with $\mathbb{Q}$-factorial terminal singularities, and assume that K_X is Cartier. Then:*

(1) $|mK_X|$ defines a birational map onto the image for any integer $m \geq d+3$.
(2) If $K_X^d \geq 2$, then $|mK_X|$ defines a birational map onto the image for any integer $m \geq d+2$.

Part (2) of Conjecture 1.1, which is [Cat22, Conjecture 0.1 b)], is due to Meng Chen. For the integers m in parts (1) and (2), [Cat22, Conjecture 0.1 a') and b')] further predicts that $|mK_X|$ embeds the canonical model of X.

Conjecture 1.1 holds for $d = 2$ by Bombieri [Bom73]. The bounds $m \geq 5$, and $m \geq 4$ when $K_X^2 \geq 2$, hold for projective surfaces X with canonical singularities such that K_X is nef and big, because the minimal resolution of X is crepant. For $d = 3$, [CCZ07, Theorem 1.1] proved that $|mK_X|$ defines a birational morphism onto the image for any integer $m \geq 5$ when X is a projective minimal Gorenstein threefold of general type with canonical singularities. Here K_X^3 is even [CCZ07, 1.4 and 2.2], so $K_X^3 \geq 2$, and the bound 5 is optimal [CCZ07, Example 1.2]. In particular, Conjecture 1.1 holds for $d = 3$ [Cat22, Introduction].

The assumption that K_X is Cartier cannot be removed. Indeed, a general hypersurface $X_{46} \subset \mathbb{P}(4, 5, 6, 7, 23)$ is a minimal threefold of general type with $\mathbb{Q}$-factorial terminal singularities, $K_{X_{46}}^3 = 1/420$, and the smallest integer m_0 such that $|mK_{X_{46}}|$ defines a birational map onto the image for any integer $m \geq m_0$ is 27 [CC10, Example 1.3] and [Cat22, Example 0.1]. See also [Rei87, p. 359] and [IF00, p. 151, No. 23].

Han proved bounds that grow linearly in d for normal varieties.

Theorem 1.2 ([Han26, Proposition 5.6 and Theorem 1.3]). *Let $C_0 = 1.77629988\ldots$ be the constant of* [Han26, Theorem 1.1] *and X a normal projective variety of dimension d. Then:*

(1) Let L be a nef and big Cartier divisor on X and D a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. Assume that $D - (C_0 d + 3\sqrt{2ed})L$ is pseudo-effective. Then $|K_X + \lceil D \rceil|$ defines a birational map onto the image.
(2) Assume that K_X is Cartier, nef, and big. Then $|mK_X|$ defines a birational map onto the image for any integer $m \geq 1 + \lceil C_0 d + 3\sqrt{2ed} \rceil$.

Here $|K_X + \lceil D \rceil|$ is the complete linear system of the rank one reflexive sheaf $\mathcal{O}_X(K_X + \lceil D \rceil)$, and e is the base of the natural logarithm.

The constant in Theorem 1.2(1) has two parts. The term $C_0 d$ is the global generation bound of [Han26, Theorem 1.1], and the term $3\sqrt{2ed}$ is the cost of the second point: the bound for r points in [Han26, Theorem 5.4] is $C_0 d + 3(r-1)\sqrt{2ed}$, while in the bound of [AS95, Corollary 0.4] each additional point adds d.

Han asked [Han26, Questions 5.10 and 5.11] whether Theorem 1.2(1) holds when $D - (d+1)L$ is big, and whether Theorem 1.2(2) holds for any integer $m \geq d+3$, and for any integer $m \geq d+2$ when $K_X^d \geq 2$. The constant $d+1$ in [Han26, Question 5.10], and in $(d+2)K_X = K_X + (d+1)K_X$ for [Han26, Question 5.11], is the constant of the global generation part of Fujita's conjecture, and the bounds $d+3$ and $d+2$ in [Han26, Question 5.11] are those of Conjecture 1.1. Thus both questions ask that the second point does not add a term such as $3\sqrt{2ed}$.

We recall the theorem of [OAI26].

Theorem 1.3 ([OAI26, Theorem 1.1 and Corollary 6.3]). *Let X be a smooth projective variety of dimension $d \geq 1$ and L an ample Cartier divisor on X. Then $K_X + mL$ is globally generated for any integer $m \geq d+1$.*

For two points, [Nug26] proved the following theorem.

Theorem 1.4 ([Nug26, Theorem A]). *Let X be a smooth projective variety of dimension $d \geq 1$ and L an ample Cartier divisor on X. Then for any integer $m \geq d+2$ and any two distinct closed points $x, y \in X$, the evaluation map*

$$H^0(X, K_X + mL) \to (K_X + mL)|_x \oplus (K_X + mL)|_y$$

is surjective.

Theorem 1.4 uses the results of [OAI26], see [Nug26, Section 1 and Appendix A].

Theorems 1.3 and 1.4 do not directly answer [Han26, Questions 5.10 and 5.11], because both assume that X is smooth and L is ample. When X is only normal, we pass to a resolution $h : X' \to X$, and h^*L is only nef and big, even if L is ample. Perturbing h^*L to an ample $\mathbb{Q}$-divisor does not help: for X smooth and L ample, the sharp constants depend on the integrality $L^{\dim V} \cdot V \geq 1$ for any closed subvariety $V \subset X$ [Nug26, Remark 7.1], and the corresponding intersection numbers of a perturbation can be smaller than 1 (for instance, on the blow-up of $\mathbb{P}^2$ at a point, with H the pullback of a line, E the exceptional curve, and $0 < \epsilon < 1$ rational, the $\mathbb{Q}$-divisor $H - \epsilon E$ is ample by [Har77, Example V.2.11.5 and Corollary V.2.18(a)] and has degree $(H - \epsilon E) \cdot (H - E) = 1 - \epsilon$ on the strict transform of a line through the point).

We now explain the choice of method in this paper. For a smooth projective variety X and an ample Cartier divisor L, the method from [AS95], in the algebraic formulation of [Kol97], to [Han26] constructs lc centers through a closed point x and cuts them down to x, and each cut adds a rational multiple of L to the boundary. If all the lc centers that are cut are smooth at x, then for any real number $\epsilon > 0$ the sum of the multiples can be taken at most $d + \epsilon$ [Han26, Section 1, (2) and the display after it] (with W_i, $d_i = \dim W_i$, b_i, and ϵ_i as in [Han26, Section 1, (2)], we have $\operatorname{mult}_x W_i = 1$ and $L^{d_i} \cdot W_i \geq 1$, so the $(i+1)$-th multiple is at most $b_i - b_{i+1} + \epsilon_i$, and since $b_0 = d$ and the last b_i is 0, the sum of the multiples is at most $d + \sum_i \epsilon_i$). Thus the excess of $C_0 d$ over d is caused by the lc centers singular at x (cf. [Hel97, Theorem 4.3] and [OAI26, Section 1]).

[OAI26] does not cut. It minimizes a mean of exponentials of the differences between the log discrepancy divided by a fixed positive real number and the orders of vanishing divided by m, over the bases of $H^0(X, mL)$ and over the valuations with center containing x that are given by weights on simple normal crossing divisors over X [OAI26, Section 4]. To show that the center of a minimizer is a closed point, it compares the growth of the images of the restriction maps $H^0(X, mL) \to H^0(V, mL|_V)$ with the first variation of the minimized quantity in directions tangent to the center V of a minimizer, and the estimate is uniform over all subvarieties of a fixed dimension, which may be singular and of unbounded degree [OAI26, Sections 1 and 5]. It then constructs from the minimizer a $\mathbb{Q}$-divisor to which Kawamata–Viehweg vanishing is applied [OAI26, Section 6]. We do not use the conclusion of Theorem 1.3, and we use the method of [OAI26].

The goal of this paper is to answer [Han26, Questions 5.10 and 5.11].

Main results. Our first main theorem considers adjoint linear systems.

Theorem 1.5. *Let X be a normal projective variety of dimension $d \geq 1$, L a nef and big Cartier divisor on X, and D a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. Assume that $D - (d+1)L$ is big. Then $|K_X + \lceil D \rceil|$ defines a birational map onto the image.*

Our second main theorem considers pluricanonical systems.

Theorem 1.6. *Let X be a normal projective variety of dimension $d \geq 1$. Assume that K_X is Cartier, nef, and big. Then:*

(1) $|mK_X|$ defines a birational map onto the image for any integer $m \geq d+3$.
(2) If $K_X^d \geq 2$, then $|mK_X|$ defines a birational map onto the image for any integer $m \geq d+2$.

Theorems 1.5 and 1.6 answer [Han26, Questions 5.10 and 5.11] in the affirmative. Since K_X is nef and big on a minimal projective variety of general type, Theorem 1.6 implies Conjecture 1.1.

Theorem 1.5 with $L = K_X$ and $D = (m-1)K_X$ implies Theorem 1.6(1). For $m = d+2$, the choice $L = K_X$ and $D = (d+1)K_X$ has $D-(d+1)L = 0$, which is not big. Thus Theorem 1.6(2) is not a special case of Theorem 1.5 (for instance, if $\mathrm{Pic}(X) = \mathbb{Z}K_X$, then no choice of L and D in Theorem 1.5 gives $|(d+2)K_X|$).

For $d = 2$, the ranges of Theorem 1.6 are those of Bombieri's bounds on surfaces with canonical singularities [Bom73]. For normal projective surfaces X with log canonical singularities such that K_X is Cartier and ample, [LR14, Theorem 5.1(ii)] proved that $|mK_X|$ defines a birational morphism onto the image for any integer $m \geq 6$. For $d = 3$, the range of Theorem 1.6(2) is that of [CCZ07, Theorem 1.1], and on the threefolds of [CCZ07, Theorem 1.1] the map defined by $|mK_X|$ is moreover a morphism.

Theorem 1.4 appeared earlier and overlaps with both main theorems. If X is smooth, L is ample, and $D = mL$ for an integer $m \geq d+2$, then Theorem 1.4 implies that $|K_X + mL|$ separates any two distinct closed points, which is stronger than the conclusion of Theorem 1.5. If X is smooth and K_X is ample, then for $m \geq d+3$, we have $mK_X = K_X + (m-1)L$ with $L = K_X$ and $m-1 \geq d+2$. Thus Theorem 1.4 implies Theorem 1.6(1) for such X. For $m = d+2$ and $L = K_X$, we have $m-1 = d+1 < d+2$, so Theorem 1.4 with $L = K_X$ does not apply to $|(d+2)K_X|$.

In Theorem 1.5, X is normal and K_X need not be $\mathbb{Q}$-Cartier, L is nef and big, D is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, the round-up is taken on D, and $D-(d+1)L$ is big. In Theorem 1.6, X is normal and K_X is Cartier, nef, and big, with no other assumption on the singularities of X.

By [Cat22, Example 1.2], for any $d \geq 1$ there exists a smooth projective variety X of dimension d with K_X ample and $K_X^d = 2$ (here X is a general hypersurface of degree $2(d+2)$ in the weighted projective space $\mathbb{P}(1,\ldots,1,d+2)$, whose first $d+1$ weights are 1, and by adjunction $\omega_X = \mathcal{O}_X(1)$, so K_X^d is the degree divided by the product of the weights, namely $2(d+2)/(d+2) = 2$) such that $|mK_X|$ defines a birational map onto the image if and only if $m \geq d+2$. Thus the bound $d+2$ in Theorem 1.6(2) is optimal. By [Cat22, Example 1.1], for any even d there exists a smooth projective variety X of dimension d with K_X ample and $K_X^d = 1$ (here X is a general hypersurface of degree $2(d+3)$ in the weighted projective space $\mathbb{P}(1,\ldots,1,2,d+3)$, whose first d weights are 1, and by adjunction $\omega_X = \mathcal{O}_X(1)$, so K_X^d is the degree divided by the product of the weights, namely $2(d+3)/(2(d+3)) = 1$) such that $|mK_X|$ defines a birational map onto the image if and only if $m \geq d+3$. Thus the bound $d+3$ in Theorem 1.6(1) is optimal for even d. See also [BPT13, Theorem 3].

If $X = \mathbb{P}^d$, L is a hyperplane in $\mathbb{P}^d$, and $D = (d+1)L$, then $K_X + \lceil D \rceil \sim 0$, and $|K_X + \lceil D \rceil|$ does not define a birational map onto the image. Thus the assumption that $D-(d+1)L$ is big cannot be weakened to $D-(d+1)L \geq 0$. For the hyperplane L in $\mathbb{P}^d$, we have $L^d = 1$. If $L^d \geq 2$ and $D-(d+1)L$ is $\mathbb{Q}$-linearly equivalent to an effective $\mathbb{Q}$-divisor, then $|K_X + \lceil D \rceil|$ defines a birational map onto the image by Proposition 6.1, applied on a projective resolution of X, and Lemma 7.1.

Sketch of the proof. Theorems 1.5 and 1.6 follow from Proposition 6.1 on a projective resolution of X and Lemma 7.1, which pushes sections forward (cf. [Han26, Proposition 5.6]). Proposition 6.1 follows from two results on the open subset U of Set-up 3.1:

(1) For any closed point $x \in U$ and any real number $t > d+1$, and for some t with $0 < t < d+1$ if $L^d \geq 2$, there exist infinitely many m such that any minimizer at x of the functional $\psi_{m,t}$ of Definition 4.1 has center $\{x\}$ (Proposition 5.1). This follows from Lemma 4.7 and the opposite bounds of Lemmas 5.4 and 5.5, the second depending on Proposition 3.2.

(2) If x is very general, then a minimizer yields a section of the divisor $K_X + M$ of Proposition 6.1, non-zero at x and zero at a given $y \in U\backslash\{x\}$ (Proposition 6.5). This follows from Kawamata–Viehweg vanishing and Lemma 6.4, which depends on Construction-Proposition 6.3.

The key step is Construction-Proposition 6.3(2): since x is very general, the numbers $v(s_i)/a(v,X,0)$ of a minimizer (v,s) at x are dominated by values of a monomial valuation w at

y, and the rearrangement inequality on a common adapted basis implies $\sum_i b_i w(s_i) > a(w, X, 0)$ for the coefficients b_i of Construction-Proposition 6.3, because $|mL|$ separates x and y. Since $\sum_i b_i w(s_i) > a(w, X, 0)$ is strict, Lemma 2.2(1) replaces w by a divisorial valuation, which yields the prime divisor E_y of Construction-Proposition 6.3(2). We only separate a very general point from the other points of U. The small degree case of Proposition 3.2, the upper bound of Lemma 4.7(1), and Construction-Proposition 6.3(2) are new, and the other steps follow [OAI26] for nef and big L.

(a) Subvarieties of small degree. The proof of [OAI26, Lemma 5.2] interpolates on k^e points of the geometric generic fiber of a general linear projection $V \to \mathbb{P}^e$, which needs $(kL)^e \cdot V \geq k^e$ for kL very ample [OAI26, Proof of Lemma 5.2, first display]. Here kL need not define an embedding, and the very ample divisor A of Lemma 2.4(2) only satisfies $A^e \cdot V \leq (kL)^e \cdot V$. The case $A^e \cdot V < k^e$ cannot be avoided: on the blow-up of $\mathbb{P}^2$ at a point, with L the pullback of a line, it occurs for the strict transform of any line through the point, for any A and k of Lemma 2.4(2) (writing $A \sim aL - bE$ with E the exceptional curve, ampleness of A implies $b > 0$ and the effectivity of $kL - A$ implies $k \geq a$, so $A \cdot V = a - b < k$).

If $A^e \cdot V \geq k^e$, then we interpolate in the embedding defined by A. If $A^e \cdot V < k^e$, then V belongs to a bounded family, so vanishing for $H + lL$ with a fixed multiple H of A is uniform in V, the leading term of $\chi(V, (H + lL)|_V)$ is $(L^e \cdot V)l^e/e!$ with $L^e \cdot V \geq 1$, and Lemma 2.4(1) removes H at the cost of a constant independent of V.

(b) The strict inequality at x. The bound $\mu_{m,t}(x) < 1$ of [OAI26, Lemma 4.2] depends on the strict inequality of [OAI26, Proposition 3.2], which is proved under the assumption that $K_X + (d+1)L$ is not generated at x. We have no such assumption. By [OAI26, Proof of Proposition 3.2, after (3.3)], the strict inequality can only fail if $L^d = 1$, and we avoid the equality case. The bound [OAI26, (3.3)] says that, for a basis of $H^0(X, mL)$ with distinct leading exponents at x, the mean multiplicity at x divided by m has lower limit at least $(L^d)^{1/d} d/(d+1)$. The assumption $t_-(L^d)^{1/d} > d + 1$ of Lemma 4.7 implies that $(L^d)^{1/d} d/(d+1) > d/t_-$, and then $\psi_{m,t}$ is smaller than 1 at a small multiple of the valuation of the exceptional divisor of the blow-up of x, so $\mu_{m,t}(x) < 1$.

If $L^d \geq 2$, then we may take $t_- < d + 1$, and the lower bound $a_* > 0$ of Lemma 4.7(2) allows us to fix $t < d + 1$ in Proposition 5.1(2) after a_* (cf. [OAI26, (5.17) and Table 1]). Since Proposition 6.5 only assumes $q > t$, the value $q = d + 1$ is allowed in Proposition 6.1. If $L^d = 1$, then the assumption of Lemma 4.7 forces $t > d + 1$, so $q > d + 1$, while the opposite bounds of Lemmas 5.4 and 5.5 allow $t < d + 1$ once $a_* > 0$. Thus Theorem 1.5 assumes that $D - (d+1)L$ is big, which yields a rational $q > d + 1$ with $D - qL$ big, and Theorem 1.6(2) assumes that $K_X^d \geq 2$.

(c) The order of the choices. The proof of Proposition 5.1 only yields infinitely many m: it runs along a sequence of degrees on which $\mu_{m,t}(x)$ tends to its lower limit, and excludes a finite set of them depending on x. Thus m cannot be fixed before x. The proof of Theorem 1.4 in [Nug26, Section 7] takes a degree good at both points [Nug26, Theorem 5.1], which depends on the convergence of the minima, proved for ample L [Nug26, Proposition 4.4]. We do not use convergence, and we choose x, then t, then m, then y: by Lemma 4.4(3), a very general x belongs to the set U_m° of Definition 4.3 for all large m, and the only condition on y is $y \in U \backslash \{x\}$. Separating x from y yields no information on base points, and suffices for birationality: otherwise any point of a non-empty open subset of U has another point in the fiber through it.

(d) The strict inequality at y. Since m depends on x, we have no minimizer at y, while [Nug26, Lemma 3.1] assumes minimizers at both points and $\mu_{m,t}(x) \geq \mu_{m,t}(y)$. The minimal values cannot yield strictness either: if $x, y \in U_m^\circ$, then $\mu_{m,t}(x) = \mu_{m,t}(y)$ by Lemma 4.4(3) and Lemma 2.2(1), together with the approximation argument in the proof of Lemma 4.6. Instead, by Lemma 4.4(3), any rational list below the numbers $v(s_i)/a(v, X, 0)$ is realized at y, with a basis and a valuation depending on the list, which only yields a non-strict inequality. Lemma 4.5 replaces the bases and valuations by one basis and one monomial valuation w at y, and Lemma 4.2(2) puts v and w

on one basis. Then equality in the rearrangement inequality implies that the sections vanishing at x are the sections vanishing at y, which contradicts Lemma 2.4(3). Since $\sum_i b_i \operatorname{div}(s_i) \equiv tL$ is determined by the minimizer at x, the point y costs no additional positivity, so the term $3\sqrt{2ed}$ of Theorem 1.2(1) does not appear.

Of the 16 numbered results of [OAI26, Sections 2–6] before [OAI26, Corollary 6.3], we cite six, namely [OAI26, Lemmas 2.1, 2.2, 2.3, 2.5, 4.1, and 6.1], and we write out seven for nef and big L, namely [OAI26, Lemmas 4.2, 4.3, 4.5, 5.2, and 6.2 and Propositions 4.4 and 5.1]. The other three are [OAI26, Theorem 2.4], which is Kawamata–Viehweg vanishing, [OAI26, Lemma 3.1], whose multiplicity bound we reprove in the proof of Lemma 4.7, and [OAI26, Proposition 3.2], which we replace as in (b).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce monomial valuations over a smooth variety, and collect basic facts on augmented base loci.

In Section 3, for a smooth projective variety X and a nef and big Cartier divisor L, we bound from below the dimension of the image of $H^0(X, mL) \to H^0(V, mL|_V)$, uniformly in the closed subvariety V of X not contained in the augmented base locus of L. In Section 4 we introduce an exponential functional of a monomial valuation and an ordered basis of $H^0(X, mL)$, and prove that minimizers exist. In Section 5 we prove that, for infinitely many m, any minimizer at a closed point x outside the augmented base locus is centered at $\{x\}$.

In Section 6 we separate a very general point from any other point of a dense open subset of X, and deduce birationality of adjoint linear systems on X. In Section 7 we prove Theorems 1.5 and 1.6.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Monomial valuations.

Let X be a variety. For a prime divisor E over X and a Cartier divisor D or a non-zero ideal $\mathfrak{a}$ on a birational model of X, we write $\operatorname{mult}_E D$ and $\operatorname{mult}_E \mathfrak{a}$ for the multiplicities along E of the pullbacks of D and $\mathfrak{a}$ to a higher model on which E is a divisor.

Definition 2.1. Let X be a smooth projective variety, $h : Y \to X$ a birational morphism from a smooth projective variety Y, Y^0 a non-empty open subset of Y, and $\sum_j E_j$ a reduced SNC divisor on Y^0 with $\operatorname{Supp} K_{Y/X}|_{Y^0} \subset \operatorname{Supp} \sum_j E_j$. For a prime divisor E on Y or on Y^0, we let $a(E, X, 0) := 1 + \operatorname{mult}_E K_{Y/X}$ be the *log discrepancy* of E. A *stratum* of $\sum_j E_j$ is an irreducible component of the intersection of some components of $\sum_j E_j$. Let W be a stratum of $\sum_j E_j$. After renumbering the E_j, W is an irreducible component of $E_1 \cap \cdots \cap E_r$ with $r \geq 1$. Let $u_1, \ldots, u_r$ be positive real numbers, $y_1, \ldots, y_r$ local equations of $E_1, \ldots, E_r$ at the generic point of W, and κ the residue field of Y at the generic point of W. The *monomial valuation* over X with weights $u_1, \ldots, u_r$ along $E_1, \ldots, E_r$ at W is the map v which sends a non-zero regular function f near the generic point of W, written as $f = \sum_\gamma c_\gamma y^\gamma$ with $c_\gamma \in \kappa$ in the completion $\kappa[[y_1, \ldots, y_r]]$ of the local ring of Y at the generic point of W, to

$$v(f) := \min \left\{ \sum_{j=1}^{r} u_j \gamma_j \;\middle|\; c_\gamma \neq 0 \right\}.$$

We let $\operatorname{center}_X v := \overline{h(W)}$ and $a(v, X, 0) := \sum_{j=1}^{r} u_j a(E_j, X, 0)$. For a Cartier divisor $D \geq 0$ on X, we let $v(D)$ be the value of v on a local equation of h^*D. For a non-zero section s of a line bundle on X, we let $v(s) := v(\operatorname{div}(s))$. If $Y^0 = Y$, then we say that v is a monomial valuation *on* $(Y, \sum_j E_j)$. For a real number $c > 0$, we let cv be the monomial valuation over X with weights $cu_1, \dots, cu_r$ along $E_1, \dots, E_r$ at W. Then $\operatorname{center}_X(cv) = \operatorname{center}_X v$, $a(cv, X, 0) = ca(v, X, 0)$, and $(cv)(D) = cv(D)$ for any Cartier divisor $D \geq 0$ on X.

For a prime divisor E over X and a real number $c > 0$, the *divisorial valuation* over X with prime divisor E and constant c is the monomial valuation over X with $r = 1$ and weight c along E at $E|_{Y^0}$, where Y is a smooth projective higher model of X on which E is a divisor and Y^0 is a neighborhood of the generic point of E in Y with $E|_{Y^0}$ smooth and $\operatorname{Supp} K_{Y/X}|_{Y^0} \subset E$. For such a valuation v, we have $a(v, X, 0) = ca(E, X, 0)$, and we let $v(D) := c \operatorname{mult}_E D$ for any Cartier divisor $D \geq 0$ on a birational model of X and $v(\mathfrak{a}) := c \operatorname{mult}_E \mathfrak{a}$ for any non-zero ideal $\mathfrak{a}$ on X.

Definition 2.1 is a local form of the definition of quasi-monomial valuations (cf. [JM12, Section 3.1] and [OAI26, Section 2.1]).

The set Y^0 may be a proper open subset of Y because Lemma 5.4 uses local parameters at a point, the zero loci of which are only local hypersurfaces.

The condition $\operatorname{Supp} K_{Y/X}|_{Y^0} \subset \operatorname{Supp} \sum_j E_j$ is used in Lemma 2.2(1): for rational weights, it ensures that the sum $\sum_j u_j a(E_j, X, 0)$ equals the log discrepancy of the corresponding divisorial valuation. Without the condition the equality can fail: if X is a smooth projective surface, h is the blow-up of a closed point, and E_1, E_2 are local curves on Y, different from the exceptional curve of h, intersecting transversally at a closed point y of the exceptional curve, then for weights $1, 1$ at y, the map v is the divisorial valuation of the exceptional divisor F of the blow-up of y, the sum is 2, and $a(F, X, 0) = 3$ since $K_{Y'/X} = E' + 2F$ on the blow-up Y' of Y at y, where E' is the strict transform of the exceptional curve of h.

Lemma 2.2 (cf. [OAI26, Section 2.1 and Lemmas 2.2 and 4.1])**.** *Let X be a smooth projective variety and v a monomial valuation over X. Then:*

1. *For any finite set $\mathcal{S}$ of non-zero sections of line bundles on X, there exist divisorial valuations v_k over X with $\operatorname{center}_X v_k = \operatorname{center}_X v$ such that $\lim_{k\to+\infty} a(v_k, X, 0) = a(v, X, 0)$ and $\lim_{k\to+\infty} v_k(s) = v(s)$ for any $s \in \mathcal{S}$.*
2. *For any closed point $z \in \operatorname{center}_X v$ and any non-zero section s of a line bundle on X, we have $v(s) \leq a(v, X, 0) \operatorname{mult}_z \operatorname{div}(s)$.*
3. *Let $D \geq 0$ be a Cartier divisor on X, $h : Y \to X$ a log resolution of (X, D), and $\sum_j E_j$ the reduced divisor on Y with support $\operatorname{Exc}(h) \cup \operatorname{Supp} h^*D$. Let w be a divisorial valuation over X such that $u_j := w(E_j) > 0$ for some j. Then the monomial valuation w' on $(Y, \sum_j E_j)$ with weights u_j along the E_j with $u_j > 0$, at the irreducible component of $\bigcap_{u_j>0} E_j$ that contains the center of w on Y, satisfies $w'(D') = w(D')$ for any Cartier divisor $D' \geq 0$ on X with $\operatorname{Supp} D' \subset \operatorname{Supp} D$, $a(w', X, 0) \leq a(w, X, 0)$, and $\operatorname{center}_X w \subset \operatorname{center}_X w'$.*

Proof. Item (1) follows by approximating the weights with rational numbers: a monomial valuation with rational weights is divisorial by a toroidal extraction [OAI26, Section 2.1, second paragraph after (2.2)], carried out on the set Y^0 of Definition 2.1 and followed by passing to a smooth projective higher model of X when Y^0 is a proper open subset of Y, and the center does not change [OAI26, paragraph after (4.1)]. Items (2) and (3) are [OAI26, Lemma 4.1] and [OAI26, Lemma 2.2]. □

2.2. **Augmented base loci.**

Definition 2.3 ([ELM+06, Section 1, Definition 1.2, and Remark 1.3])**.** Let X be a smooth projective variety and D a $\mathbb{Q}$-divisor on X. The *stable base locus* of D is $\mathbf{B}(D) := \bigcap_m \operatorname{Bs}|mD|$, the intersection being over all positive integers m such that mD is integral. The *augmented base locus* of D is $\mathbf{B}_+(D) := \bigcap_A \mathbf{B}(D - A)$, the intersection being over all ample $\mathbb{Q}$-divisors A on X.

If L is a big $\mathbb{Q}$-divisor on X, then $\mathbf{B}_+(L) \neq X$ [ELM+06, Example 1.7]. If L is only nef and big, then mL need not be base point free or very ample, and Lemma 2.4 is what we use instead on $X \backslash \mathbf{B}_+(L)$.

Lemma 2.4. *Let X be a smooth projective variety, L a nef and big Cartier divisor on X, and $U := X \backslash \mathbf{B}_+(L)$. Then:*

1. *For any globally generated ample Cartier divisor A on X, there exists a positive integer k with $\mathrm{Bs}|kL - A| \subset \mathbf{B}_+(L)$.*
2. *There exist a very ample Cartier divisor A on X and a positive integer k such that $\mathrm{Bs}|kL - A| \subset \mathbf{B}_+(L)$ and $A + lL$ is very ample for any integer $l \geq 0$.*
3. *For k as in (2) and any integer $l \geq k$, $|lL|$ has no base point in U and separates any two distinct closed points of U.*

Proof. We prove (1). Let A be a globally generated ample Cartier divisor on X. By [ELM+06, Proposition 1.5] applied to the ample $\mathbb{Q}$-divisor $\frac{1}{n}A$ for n large, there exists a positive integer n with $\mathbf{B}_+(L) = \mathbf{B}(L - \frac{1}{n}A)$. Since $\mathrm{Bs}|lD| \subset \mathrm{Bs}|D|$ for any Cartier divisor D on X and any positive integer l, Definition 2.3 implies that $\mathbf{B}(L - \frac{1}{n}A) = \mathbf{B}(nL - A)$. Thus

$$\mathbf{B}_+(L) = \mathbf{B}(nL - A).$$

By [Laz04, Proposition 2.1.21], there exists a positive integer j with $\mathrm{Bs}|j(nL - A)| = \mathbf{B}(nL - A)$. Let $k := jn$. For any closed point $x \notin \mathbf{B}_+(L)$, there exist $\sigma_1 \in H^0(X, j(nL - A))$ and $\sigma_2 \in H^0(X, (j-1)A)$ not vanishing at x. Since

$$j(nL - A) + (j-1)A = kL - A,$$

$\sigma_1\sigma_2$ is a section of $kL - A$ not vanishing at x. Thus $\mathrm{Bs}|kL - A| \subset \mathbf{B}_+(L)$.

We prove (2). Let A_0 be a very ample Cartier divisor on X. Since L is nef and the cohomology of a coherent sheaf on X vanishes in degrees larger than $\dim X$, Fujita vanishing [Laz04, Theorem 1.4.35] implies that there exists a positive integer a with

$$H^i(X, (a-i)A_0 + lL) = 0 \text{ for any integers } i > 0 \text{ and } l \geq 0.$$

Thus $aA_0 + lL$ is 0-regular with respect to A_0 for any integer $l \geq 0$, hence globally generated [Laz04, Theorem 1.8.5]. Let $A := (a+1)A_0$. For any integer $l \geq 0$,

$$A + lL = A_0 + (aA_0 + lL)$$

is the sum of a very ample divisor and a globally generated divisor, hence very ample [Har77, Exercise II.7.5(d)]. In particular, A is very ample, hence a globally generated ample Cartier divisor. By (1), there exists a positive integer k with $\mathrm{Bs}|kL - A| \subset \mathbf{B}_+(L)$.

We prove (3). Let A and k be as in (2), $l \geq k$ an integer, and x, x' closed points of U, possibly equal. By (2), the sections of $kL - A$ vanishing at x and the sections of $kL - A$ vanishing at x' form proper linear subspaces of $H^0(X, kL - A)$. Since a complex vector space is not a finite union of proper linear subspaces, there exists $\sigma \in H^0(X, kL - A)$ not vanishing at x or x'. Since $A + (l-k)L$ is very ample by (2), the linear subsystem

$$\sigma \cdot H^0(X, A + (l-k)L) \subset H^0(X, lL)$$

has no base point at x or x', and separates x and x' if $x \neq x'$. Thus (3) holds. □

3. Uniform estimates for restriction maps

In this section, for a nef and big Cartier divisor L on a smooth projective variety X, we bound from below the dimension of the image of $H^0(X, lL) \to H^0(V, lL|_V)$, uniformly in the closed subvariety V of X not contained in $\mathbf{B}_+(L)$, and we construct sections of a fixed degree that give local parameters on V at a smooth point of V outside $\mathbf{B}_+(L)$ (Proposition 3.2). In Section 5 the varieties V are centers of valuations and vary with the degree m, so the constants cannot depend on V. We also need the leading coefficient to be exactly one: a lower bound $c_0 l - c_e$ with $c_0 < 1$ would only imply Theorem 1.5 with $(d+1)/c_0$ in place of $d+1$, since the lower bound

$e/(d+1)$ for $\liminf T_m$ in the proof of Proposition 5.1, with the exponential factor discarded, would become $ec_0/(d+1)$.

Set-up 3.1. Let X, L, U, R_m, N_m, and $I_V(m)$ be as follows.

(1) X is a smooth projective variety of dimension $d \geq 1$.
(2) L is a nef and big Cartier divisor on X, and $U := X \backslash \mathbf{B}_+(L)$.
(3) For any positive integer m, $R_m := H^0(X, mL)$ and $N_m := \dim R_m$.
(4) For any closed subvariety V of X and any positive integer m, $I_V(m)$ is the dimension of the image of the restriction map $R_m \to H^0(V, mL|_V)$.

We only consider closed subvarieties V with $V \cap U \neq \emptyset$ because the lower bound on $I_V(l)$ can fail for V contained in $\mathbf{B}_+(L)$: if X is the blow-up of $\mathbb{P}^2$ at a point, L is the pullback of $\mathcal{O}_{\mathbb{P}^2}(1)$, and V is the exceptional curve, then $I_V(l) = 1$ for any positive integer l, because $L|_V$ is trivial and $|lL|$ has members not containing V. Here $V \subset \mathbf{B}_+(L)$, since $(L - A) \cdot V < 0$ for any ample $\mathbb{Q}$-divisor A on X (Definition 2.3). The centers in Section 5 contain a point of U.

Unlike in [OAI26, Lemma 5.2], a multiple of L need not define an embedding, so we split according to the degree of V with respect to an auxiliary very ample divisor (see (a) in the sketch of the proof in Section 1).

Proposition 3.2 (cf. [OAI26, Lemma 5.2])**.** *Notation and conditions as in Set-up 3.1. Then:*

(1) For any integer e with $1 \leq e \leq d-1$, there exists a positive integer c_e such that

$$(e! I_V(l))^{1/e} \geq l - c_e$$

for any integer $l \geq c_e$ and any closed subvariety V of X with $\dim V = e$ and $V \cap U \neq \emptyset$.

(2) There exists a positive integer k such that for any closed subvariety V of X with $\dim V = e \geq 1$ and any smooth point y of V with $y \in U$, there exist $\tau_0, \dots, \tau_e \in R_k$ with $\tau_0(y) \neq 0$ and $\tau_i(y) = 0$ for any $i \geq 1$ such that $(\tau_1/\tau_0)|_V, \dots, (\tau_e/\tau_0)|_V$ form a regular system of parameters of $\mathcal{O}_{V,y}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix an auxiliary very ample divisor A and a positive integer k, and we prove that $L^e \cdot V \geq 1$ for any closed subvariety V of X with $\dim V = e \geq 1$ and $V \cap U \neq \emptyset$.

By Lemma 2.4(2), there exist a very ample Cartier divisor A on X and a positive integer k such that $\mathrm{Bs}|kL - A| \subset \mathbf{B}_+(L)$ and $A + lL$ is very ample for any integer $l \geq 0$. Neither A nor k depends on e.

Let V be a closed subvariety of X with $\dim V = e \geq 1$ and $V \cap U \neq \emptyset$. Since $\mathrm{Bs}|kL - A| \subset \mathbf{B}_+(L)$, there exists $\sigma_V \in H^0(X, kL - A)$ with $\sigma_V|_V \neq 0$. Thus $kL|_V \sim A|_V + G$, where $G \geq 0$ is a Cartier divisor on V. Since L is nef and A is ample,

$$(kL)^i \cdot A^{e-i} \cdot V - (kL)^{i-1} \cdot A^{e-i+1} \cdot V = (kL|_V)^{i-1} \cdot (A|_V)^{e-i} \cdot G \geq 0$$

for any integer i with $1 \leq i \leq e$, since G is effective [Laz04, Example 1.4.16]. Thus

$$(kL)^e \cdot V \geq (kL)^{e-1} \cdot A \cdot V \geq \cdots \geq A^e \cdot V > 0.$$

Since L is Cartier, $L^e \cdot V$ is an integer, so

$$L^e \cdot V \geq 1. \tag{3.1}$$

Step 2. In this step we prove the estimate of (1) by interpolation for the closed subvarieties V of large degree with respect to A.

Let e be an integer with $1 \leq e \leq d-1$ and let $n_0 := k^e$. Let V be a closed subvariety of X with $\dim V = e$, $V \cap U \neq \emptyset$, and $A^e \cdot V \geq n_0$. Let $s_0, \dots, s_e$ be general elements of $H^0(X, A)$. Then $s_0|_V, \dots, s_e|_V$ have no common zero, since $|A|$ is base point free and $\dim V = e$, so they define a morphism $\pi : V \to \mathbb{P}^e$ with $\pi^* \mathcal{O}_{\mathbb{P}^e}(1) = \mathcal{O}_V(A)$. Since $\mathcal{O}_V(A)$ is ample, any fiber of π is a finite set, so π is finite. By [Ful98, Proposition 2.3(c)], $\deg \pi = A^e \cdot V$. By [Har77, Corollary III.10.7], applied to the smooth locus of V and after removing the closed subset $\pi(\mathrm{Sing}\, V)$ of $\mathbb{P}^e$, which is

proper since π is finite and $\dim \operatorname{Sing} V < e$, π is étale over a neighborhood of a general closed point ξ of $\mathbb{P}^e$. Thus $\pi^{-1}(\xi)$ consists of $A^e \cdot V$ distinct points, and s_0 does not vanish at any of them.

Let $y_1, \dots, y_{n_0}$ be distinct points of $\pi^{-1}(\xi)$. Since A is very ample, for any $i \neq i'$ there exists $\lambda_{i,i'} \in H^0(X, A)$ with $\lambda_{i,i'}(y_i) \neq 0$ and $\lambda_{i,i'}(y_{i'}) = 0$. Let

$$F_i := s_0 \prod_{i' \neq i} \lambda_{i,i'} \in H^0(X, n_0 A).$$

Then $F_i(y_i) \neq 0$ and $F_i(y_{i'}) = 0$ for any $i \neq i'$, so the matrix of values $(F_i(y_{i'}))_{i,i'}$ is invertible. Let

$$\phi : \mathcal{O}_{\mathbb{P}^e}^{\oplus n_0} \to \pi_* \mathcal{O}_V(n_0 A)$$

be the map defined by the sections $F_1|_V, \dots, F_{n_0}|_V$. Since π is étale over a neighborhood of ξ, $\pi_* \mathcal{O}_V(n_0 A)$ is locally free near ξ, and the fiber of ϕ at ξ is injective because the matrix $(F_i(y_{i'}))_{i,i'}$ is invertible. Thus ϕ is injective near ξ. Since $\mathcal{O}_{\mathbb{P}^e}^{\oplus n_0}$ is torsion-free, ϕ is injective.

Let $j \geq n_0$ be an integer. Twisting ϕ by $\mathcal{O}_{\mathbb{P}^e}(j - n_0)$ and taking global sections, we obtain an injective map

$$\phi_j : H^0(\mathbb{P}^e, \mathcal{O}_{\mathbb{P}^e}(j - n_0))^{\oplus n_0} \to H^0(V, jA|_V).$$

Since the global sections of $\mathcal{O}_{\mathbb{P}^e}(j - n_0)$ are the polynomials of degree $j - n_0$ in $s_0, \dots, s_e$, the image of ϕ_j is contained in the image of the restriction map $H^0(X, jA) \to H^0(V, jA|_V)$. Multiplication by σ_V^j maps $H^0(X, jA)$ to R_{jk}, and it is injective on sections over V because V is integral and $\sigma_V|_V \neq 0$. Thus

$$I_V(jk) \geq n_0 \binom{j - n_0 + e}{e}.$$

Since $e!\binom{j-n_0+e}{e} = \prod_{t=1}^{e}(j - n_0 + t) \geq (j - n_0)^e$ and $n_0^{1/e} = k$,

$$(e! I_V(jk))^{1/e} \geq k(j - n_0). \tag{3.2}$$

Let $c' := k(n_0 + 2)$ and let $l \geq c'$ be an integer. Write $l = jk + k + r$, where j, r are integers with $0 \leq r < k$. Then $j \geq n_0$. Since $k + r \geq k$ and $V \cap U \neq \emptyset$, Lemma 2.4(3) implies that there exists $\tau \in R_{k+r}$ with $\tau|_V \neq 0$. Multiplication by τ maps the image of R_{jk} in $H^0(V, jkL|_V)$ injectively into the image of R_l in $H^0(V, lL|_V)$, so $I_V(l) \geq I_V(jk)$. By (3.2),

$$(e! I_V(l))^{1/e} \geq k(j - n_0) = l - k - r - kn_0 \geq l - c'. \tag{3.3}$$

Step 3. In this step we use that the closed subvarieties of small degree with respect to A form a bounded family, and we obtain vanishing for one divisor H that does not depend on V.

Let $\mathcal{V}$ be the set of closed subvarieties V of X with $\dim V = e$, $V \cap U \neq \emptyset$, and $A^e \cdot V < n_0$. We may assume that $\mathcal{V} \neq \emptyset$, since otherwise there is nothing to prove in this step. Let $X \subset \mathbb{P}^q$ be the embedding defined by $|A|$. For any $V \in \mathcal{V}$, the homogeneous coordinate ring of $V \subset \mathbb{P}^q$ is a reduced equidimensional standard graded $\mathbb{C}$-algebra of dimension $e + 1$ and multiplicity $A^e \cdot V < n_0$. Thus only finitely many Hilbert polynomials occur for $V \in \mathcal{V}$ (Kleiman, see [RTV04, Theorem 3.4]). Let S be the finite disjoint union of the Hilbert schemes [Gro61], see [FGI+05, Theorem 5.14], parametrizing closed subschemes of $X \subset \mathbb{P}^q$ whose Hilbert polynomial occurs for some $V \in \mathcal{V}$. Let $\mathcal{Z} \subset X \times S$ be the universal family and $\mathcal{J}$ the ideal sheaf of $\mathcal{Z}$ in $\mathcal{O}_{X \times S}$. Any $V \in \mathcal{V}$ is the fiber $\mathcal{Z}_\zeta$ of $\mathcal{Z}$ over a closed point ζ of S. Since $\mathcal{O}_{X \times S}$ and $\mathcal{O}_{\mathcal{Z}}$ are flat over S, $\mathcal{J}$ is flat over S [Har77, Proposition III.9.1A(e)].

For a Cartier divisor D on X, let $\mathcal{O}(D)$ be the pullback of $\mathcal{O}_X(D)$ to $X \times S$. Cover S by finitely many affine open subsets and let T be one of them. For any coherent sheaf $\mathcal{F}$ on $X \times T$ flat over T, there exist a positive integer a, a finite direct sum $\mathcal{E}$ of copies of $\mathcal{O}(-aA)|_{X \times T}$, and a surjective map $\mathcal{E} \to \mathcal{F}$ whose kernel is coherent and flat over T. Such a map exists by [Laz04, Theorem 1.7.6(iii)] since T is affine, and its kernel is flat over T by [Har77, Proposition III.9.1A(e)]

since $\mathcal{E}$ and $\mathcal{F}$ are flat over T. Taking such a map d times, first onto $\mathcal{J}|_{X\times T}$ and then onto each kernel, we obtain an exact sequence

$$0 \to \mathcal{K} \to \mathcal{E}_{d-1} \to \cdots \to \mathcal{E}_0 \to \mathcal{J}|_{X\times T} \to 0,$$

where $\mathcal{E}_u$ is a finite direct sum of copies of $\mathcal{O}(-a_{u,T}A)|_{X\times T}$ for a positive integer $a_{u,T}$, and the kernel of any map in the sequence is flat over T. Let $V \in \mathcal{V}$ with $V = \mathcal{Z}_\zeta$ and $\zeta \in T$. Since the kernels are flat over T, the sequence stays exact after restriction to $X \times \{\zeta\}$. Since $\mathcal{O}_\mathcal{Z}$ is flat over S, the restriction of $\mathcal{J}$ to $X \times \{\zeta\}$ is $\mathcal{I}_V$. Thus we obtain an exact sequence

$$0 \to \mathcal{K}_V \to \mathcal{E}_{d-1,V} \to \cdots \to \mathcal{E}_{0,V} \to \mathcal{I}_V \to 0 \tag{3.4}$$

on X, where $\mathcal{E}_{u,V}$ is a finite direct sum of copies of $\mathcal{O}_X(-a_{u,T}A)$. Only finitely many integers $a_{u,T}$ occur, since the cover of S by the affine open subsets T is finite.

By [Laz04, Theorem 1.4.35], there exists a positive integer m_0 with $H^i(X, \mathcal{O}_X(mA + N)) = 0$ for any $i > 0$, any integer $m \geq m_0$, and any nef Cartier divisor N on X. Let $b \geq 2$ be an integer such that $b \geq m_0$ and $b - a_{u,T} \geq m_0$ for any $a_{u,T}$ above, and let $H := bA$. Since lL is nef for any integer $l \geq 0$,

$$H^i(X, \mathcal{O}_X(H + lL)) = 0 \quad \text{and} \quad H^i(X, \mathcal{E}_{u,V} \otimes \mathcal{O}_X(H + lL)) = 0 \tag{3.5}$$

for any $i > 0$, any integer $l \geq 0$, any u, and any $V \in \mathcal{V}$. Tensoring (3.4) with $\mathcal{O}_X(H + lL)$ and shifting dimensions by (3.5), we obtain

$$H^i(X, \mathcal{I}_V \otimes \mathcal{O}_X(H + lL)) \cong H^{i+d}(X, \mathcal{K}_V \otimes \mathcal{O}_X(H + lL)) = 0 \tag{3.6}$$

for any $i > 0$, since $i + d > \dim X$. By (3.5), (3.6), and the exact sequence

$$0 \to \mathcal{I}_V \otimes \mathcal{O}_X(H + lL) \to \mathcal{O}_X(H + lL) \to \mathcal{O}_V(H + lL) \to 0,$$

for any $V \in \mathcal{V}$ and any integer $l \geq 0$, the restriction map

$$H^0(X, H + lL) \to H^0(V, (H + lL)|_V) \tag{3.7}$$

is surjective, and

$$H^i(V, (H + lL)|_V) = 0 \quad \text{for any } i > 0. \tag{3.8}$$

Step 4. In this step we prove the estimate of (1) for any $V \in \mathcal{V}$.

Let $V \in \mathcal{V}$ and let $P_V(l) := \chi(V, (H + lL)|_V)$. By [Kol96, Theorem VI.2.13], applied to the coherent sheaf $\mathcal{O}_V(H)$ and the line bundle $\mathcal{O}_V(L)$, P_V is a polynomial of degree at most e. By [Laz04, Theorem 1.1.24], applied to the coherent sheaf $\mathcal{O}_V(H)$ of rank one, the coefficient of l^e in P_V is $L^e \cdot V/e!$.

We show that only finitely many polynomials P_V occur for $V \in \mathcal{V}$. Let $S_1, \ldots, S_g$ be the irreducible components of S with the reduced structure, and let $\mathcal{Z}_i := \mathcal{Z} \times_S S_i \subset X \times S_i$. Then $\mathcal{Z}_i$ is flat over S_i. By Lemma 2.4(2), $A + lL$ is very ample for any integer $l \geq 0$. Thus $H + lL = (A + lL) + (b - 1)A$ is very ample [Har77, Exercise II.7.5(d)]. For any integer l with $0 \leq l \leq e$, $|H + lL|$ embeds X into a projective space $\mathbb{P}^{q_l}$, so $\mathcal{Z}_i$ is a closed subscheme of $\mathbb{P}^{q_l} \times S_i$ flat over the integral noetherian scheme S_i. By [Har77, Theorem III.9.9], the Hilbert polynomial of a fiber of $\mathcal{Z}_i$ over S_i does not depend on the fiber, and we denote it by $Q_{i,l}$. If V is a fiber of $\mathcal{Z}_i$, then $Q_{i,l}(1) = \chi(V, (H + lL)|_V)$ [Har77, Exercise III.5.2]. Thus $P_V(l)$ takes finitely many values for any integer l with $0 \leq l \leq e$. Since $\deg P_V \leq e$, interpolation implies that only finitely many polynomials P_V occur.

By (3.1), the coefficient of l^e in $e!P_V$ is at least 1. Let M_0 be a positive integer that is at least the sum of the absolute values of the coefficients of $1, l, \ldots, l^{e-1}$ in $e!P_V$ for any $V \in \mathcal{V}$. Then

$$e!P_V(l) \geq l^e - M_0 l^{e-1} = l^{e-1}(l - M_0) \geq (l - M_0)^e$$

for any integer $l \geq M_0$. By (3.8), $h^0(V, (H + lL)|_V) = P_V(l)$, so

$$(e!h^0(V, (H + lL)|_V))^{1/e} \geq l - M_0 \tag{3.9}$$

for any integer $l \geq M_0$.

Since H is very ample, Lemma 2.4(1) implies that there exists a positive integer k' with $\mathrm{Bs}|k'L-H| \subset \mathbf{B}_+(L)$. Since $V \cap U \neq \emptyset$, there exists $f_V \in H^0(X, k'L-H)$ with $f_V|_V \neq 0$. Let $l \geq 0$ be an integer. By the surjectivity of (3.7), any section of $(H+lL)|_V$ is the restriction of an element of $H^0(X, H+lL)$. Multiplying such an element by f_V, we obtain an element of $R_{k'+l}$. Since V is integral and $f_V|_V \neq 0$,

$$I_V(k'+l) \geq h^0(V, (H+lL)|_V).$$

Let $c'' := k' + M_0$. By (3.9),

$$(e!I_V(l))^{1/e} \geq l - c'' \tag{3.10}$$

for any integer $l \geq c''$ and any $V \in \mathcal{V}$.

Step 5. In this step we conclude the proof.

We first prove (1). If $\mathcal{V} = \emptyset$, let $c_e := c'$. Otherwise let $c_e := \max\{c', c''\}$. Then (1) follows from (3.3) and (3.10).

We now prove (2), with k as in **Step 1**. Let V be a closed subvariety of X with $\dim V = e \geq 1$ and y a smooth point of V with $y \in U$. Since $\mathrm{Bs}|kL-A| \subset \mathbf{B}_+(L)$, there exists $\sigma \in H^0(X, kL-A)$ with $\sigma(y) \neq 0$. The embedding of X defined by $|A|$ induces a surjective map from the cotangent space of the projective space at y onto the cotangent space of V at y. Thus there exist $s_0, \ldots, s_e \in H^0(X, A)$ with $s_0(y) \neq 0$ and $s_i(y) = 0$ for any $i \geq 1$ such that $(s_1/s_0)|_V, \ldots, (s_e/s_0)|_V$ form a regular system of parameters of $\mathcal{O}_{V,y}$. Let $\tau_i := \sigma s_i \in R_k$ for any i. Then $\tau_0(y) \neq 0$, $\tau_i(y) = 0$ for any $i \geq 1$, and $\tau_i/\tau_0 = s_i/s_0$ for any i. □

4. Exponential minimization

In this section, we prove the existence of minimizers of the exponential functional of [OAI26, Section 4] (Lemma 4.7(3)), and we prove the properties of the functional used in Section 5 and Section 6.

4.1. The exponential functional.

In this subsection, we define the exponential functional, minimizers, and good pairs (m,t) (cf. [Nug26, Sections 2.3 and 5]), and we prove the properties of adapted bases (cf. [Nug26, Section 2.2]). Subsections 4.1 and 4.2 only involve the space R_m and do not use the positivity of L.

Definition 4.1. Notation and conditions as in Set-up 3.1. Let m be a positive integer with $N_m > 0$ and t a positive real number. For a monomial valuation v over X and an ordered basis $s = (s_1, \ldots, s_{N_m})$ of R_m, we let

$$\psi_{m,t}(v,s) := \frac{1}{N_m} \sum_{i=1}^{N_m} \exp\left(\frac{a(v,X,0)}{t} - \frac{v(s_i)}{m}\right).$$

For a closed point $z \in X$, we let $\mu_{m,t}(z)$ be the infimum of $\psi_{m,t}(v,s)$ over all monomial valuations v over X with $z \in \mathrm{center}_X v$ and all ordered bases s of R_m. We say that (v,s) is a *minimizer of* $\psi_{m,t}$ *at* z if $z \in \mathrm{center}_X v$ and $\psi_{m,t}(v,s) = \mu_{m,t}(z)$. We say that (m,t) is *good at* z if $\psi_{m,t}$ has a minimizer at z and $\mathrm{center}_X v = \{z\}$ for any minimizer (v,s) of $\psi_{m,t}$ at z. We say that an ordered basis s of R_m is *adapted to* v if for any real number λ, any non-zero $\sigma \in R_m$ with $v(\sigma) \geq \lambda$ is a linear combination of the s_i with $v(s_i) \geq \lambda$.

The functional $\psi_{m,t}$ links three parts of the proof (cf. [OAI26, Section 1]): Lemma 4.7 and Construction-Proposition 6.3(1) vary the scale of v, Lemma 5.4 varies v in directions tangent to its center, and Construction-Proposition 6.3 uses coefficients proportional to the derivatives of $\psi_{m,t}$ in the numbers $v(s_i)$. For (m,t) to be good at z, we require $\mathrm{center}_X v = \{z\}$ for any minimizer (v,s) at z, not only for some (cf. the paragraph after [OAI26, Proposition 5.1]), since otherwise a minimizer may be defined on a stratum with positive-dimensional image on X, and we do not obtain Lemma 6.4(2). The infimum $\mu_{m,t}(z)$ runs over all v with $z \in \mathrm{center}_X v$, since weights may become zero and the center may grow in a limit (Lemma 4.5(1)).

We use the exponential of the exponents of $\psi_{m,t}$ and not their linear average $a(v,X,0)/t - S_m(v,s)$, where $S_m(v,s) := \sum_i v(s_i)/(mN_m)$, for two reasons. First, the linear average is homogeneous of degree one in v and leaves only the ratio $a(v,X,0)/S_m(v,s)$, the analogues of Lemmas 5.4 and 5.5 contradict each other at a minimizer of the ratio only if the minimum exceeds $d+1$, and, by (4.4) for a divisorial valuation with prime divisor the exceptional divisor of the blow-up of z, we have $\limsup_m \inf_{v,s} a(v,X,0)/S_m(v,s) \leq (d+1)/(L^d)^{1/d}$, which is at most $d+1$, so the linear average does not exclude a positive-dimensional center. Second, the linear average has equal coefficients, while the strictness in Construction-Proposition 6.3(2) needs coefficients strictly decreasing in $v(s_i)$.

Lemma 4.2 (cf. [Nug26, Lemma 2.1])**.** *Notation and conditions as in Set-up 3.1. Let m be a positive integer with $N_m > 0$ and v a monomial valuation over X. Then:*

(1) For any ordered basis s of R_m adapted to v and any ordered basis s' of R_m, there exists a permutation ρ of $\{1,\dots,N_m\}$ such that $v(s_{\rho(i)}) \geq v(s'_i)$ for any i. In particular, $\psi_{m,t}(v,s) \leq \psi_{m,t}(v,s')$ for any positive real number t.

(2) For any monomial valuation w over X, there exists an ordered basis of R_m adapted to both v and w.

Proof. For any real number λ, let $R^\lambda_{m,v}$ be the set consisting of 0 and of the non-zero $\sigma \in R_m$ with $v(\sigma) \geq \lambda$. Since $v(\sigma+\sigma') \geq \min(v(\sigma), v(\sigma'))$ for any non-zero $\sigma, \sigma' \in R_m$ with $\sigma + \sigma' \neq 0$ [JM12, Proposition 3.1], $R^\lambda_{m,v}$ is a linear subspace of R_m.

(1) Let ρ_1 and ρ_2 be permutations of $\{1,\dots,N_m\}$ such that

$$v(s_{\rho_1(1)}) \geq \cdots \geq v(s_{\rho_1(N_m)}) \text{ and } v(s'_{\rho_2(1)}) \geq \cdots \geq v(s'_{\rho_2(N_m)}).$$

Fix i and let $\lambda := v(s'_{\rho_2(i)})$. Then $s'_{\rho_2(1)}, \dots, s'_{\rho_2(i)}$ are linearly independent elements of $R^\lambda_{m,v}$, and thus $\dim R^\lambda_{m,v} \geq i$. Since s is adapted to v, $R^\lambda_{m,v}$ is spanned by the s_l with $v(s_l) \geq \lambda$. Thus at least i of the s_l satisfy $v(s_l) \geq \lambda$, and hence $v(s_{\rho_1(i)}) \geq \lambda = v(s'_{\rho_2(i)})$. Let $\rho := \rho_1 \circ \rho_2^{-1}$. Then $v(s_{\rho(i)}) \geq v(s'_i)$ for any i. Since any term of $\psi_{m,t}$ is strictly decreasing in $v(s_i)$, we have

$$\psi_{m,t}(v,s) = \frac{1}{N_m}\sum_{i=1}^{N_m} \exp\left(\frac{a(v,X,0)}{t} - \frac{v(s_{\rho(i)})}{m}\right) \leq \psi_{m,t}(v,s').$$

(2) Since $\dim R_m < +\infty$, the decreasing families $(R^\lambda_{m,v})_\lambda$ and $(R^\lambda_{m,w})_\lambda$ consist of finitely many distinct subspaces of R_m. By [Nug26, Lemma 2.1], there exists a basis of R_m such that any $R^\lambda_{m,v}$ and any $R^\lambda_{m,w}$ is spanned by the basis elements it contains. Any ordering of the basis is adapted to both v and w. Taking $w = v$, we obtain an ordered basis of R_m adapted to v. □

4.2. **Realizable lists.** In this subsection, we introduce the lists realized by a closed point of X (cf. [OAI26, Lemma 4.3]), and we prove the closedness, the passage to limits, and the specialization of such lists. Lemma 4.6 is used in Construction-Proposition 5.2(1), and Lemma 4.4(3) and Lemma 4.5 pass the numbers $v(s_i)/a(v,X,0)$ of a minimizer at a very general closed point x to a second closed point y in Section 6.

Definition 4.3. Notation and conditions as in Set-up 3.1. Let m be a positive integer with $N_m > 0$. A *decreasing list* is a sequence of rational numbers $\alpha_1 \geq \cdots \geq \alpha_{N_m} \geq 0$. We say that a closed point $z \in X$ *realizes* α *in degree* m if there exist an ordered basis s of R_m and a divisorial valuation w over X with $z \in \operatorname{center}_X w$ such that $w(s_i) \geq \alpha_i a(w,X,0)$ for any i. We let $Z_{m,\alpha}$ be the set of closed points of X which realize α in degree m, and we let U°_m be the set of closed points of U which do not belong to any $Z_{m,\alpha} \neq X$.

The definition of a realized list keeps only the normalized orders of the sections and forgets the basis and the valuation, and the list α is exactly the data which passes from one closed point of X to another. The lists are rational because Lemma 4.4(3) needs countably many of them, and they are decreasing because, after reordering the basis, the condition $w(s_i) \geq \alpha_i a(w,X,0)$

for any i then only depends on the flag spanned by s. We use divisorial valuations because the condition is an inequality for a log canonical threshold, which is computed by prime divisors on a log resolution, and thus a minimizer, as a limit of divisorial valuations, is only a monomial valuation.

Lemma 4.4 (cf. [OAI26, Lemma 4.3]). *Notation and conditions as in Set-up 3.1. Let m be a positive integer with $N_m > 0$ and α a decreasing list. Then:*

(1) $Z_{m,\alpha}$ is a Zariski closed subset of X.
(2) For any closed point $z \in X$, the set of complete flags $F_1 \subset \cdots \subset F_{N_m} = R_m$ with $\dim F_i = i$ for which there exists a divisorial valuation w over X with $z \in \operatorname{center}_X w$ and $w(\sigma) \geq \alpha_i a(w, X, 0)$ for any i and any non-zero $\sigma \in F_i$ is a Zariski closed subset of the complete flag variety of R_m.
(3) U_m° is the complement in U of countably many proper Zariski closed subsets of X, and any decreasing list realized in degree m by a closed point of U_m° is realized in degree m by any closed point of X.

Proof. Let $\mathcal{F}$ be the complete flag variety of R_m, and let Λ_α be the set of pairs $(z, F_\bullet)$ of a closed point $z \in X$ and a flag $F_\bullet$ in the set of (2) for z.

Let w be a divisorial valuation over X, $F_\bullet$ a flag in $\mathcal{F}$, and s an ordered basis of R_m such that $s_1, \ldots, s_i$ span F_i for any i. If $w(s_i) \geq \alpha_i a(w, X, 0)$ for any i, then for any i and any non-zero $\sigma \in F_i$, we have

$$w(\sigma) \geq \min_{l \leq i} w(s_l) \geq \min_{l \leq i} \alpha_l a(w, X, 0) = \alpha_i a(w, X, 0)$$

by [JM12, Proposition 3.1], because α is decreasing. The converse holds because $s_i \in F_i$. Any ordered basis of R_m spans a flag in this way, and any flag in $\mathcal{F}$ is spanned in this way by an ordered basis. Thus $Z_{m,\alpha}$ is the image of Λ_α under the projection $X \times \mathcal{F} \to X$, and the set in (2) is the fiber of Λ_α over z. Since $\mathcal{F}$ is proper, (1) and (2) follow once we prove that Λ_α is a Zariski closed subset of $X \times \mathcal{F}$.

If $\alpha = 0$, then $\Lambda_\alpha = X \times \mathcal{F}$, since a divisorial valuation over X whose center contains z exists for any closed point $z \in X$. Assume that $\alpha \neq 0$.

Let $\mathcal{V}_1 \subset \cdots \subset \mathcal{V}_{N_m}$ be the universal subbundles on $\mathcal{F}$, with $\operatorname{rank} \mathcal{V}_i = i$. For any i, let $\mathfrak{b}_i$ be the ideal on $X \times \mathcal{F}$ locally generated by the functions ω/θ, where ω runs over a local basis of $\mathcal{V}_i$ and θ is a local generator of $\mathcal{O}_X(mL)$. For a flag $F_\bullet$ in $\mathcal{F}$, the restrictions of the functions ω/θ to $X \times \{F_\bullet\}$ generate the base ideal $\mathfrak{b}_{i,F_\bullet}$ of F_i on X, and

$$w(\mathfrak{b}_{i,F_\bullet}) = \min\{w(\sigma) \mid 0 \neq \sigma \in F_i\}$$

for any divisorial valuation w over X, since $\mathfrak{b}_{i,F_\bullet}$ is generated by the quotients by θ of a basis of F_i and any $\sigma \in F_i$ is a linear combination of that basis [JM12, Proposition 3.1]. Let n be a positive integer such that n/α_i is an integer for any i with $\alpha_i > 0$, and let

$$\mathfrak{a}_\alpha := \sum_{\alpha_i > 0} \mathfrak{b}_i^{n/\alpha_i} \text{ and } \mathfrak{a}_{\alpha,F_\bullet} := \sum_{\alpha_i > 0} \mathfrak{b}_{i,F_\bullet}^{n/\alpha_i}.$$

Let E be a prime divisor over X, c a positive real number, and w the divisorial valuation over X with prime divisor E and constant c. Since

$$w(\mathfrak{a}_{\alpha,F_\bullet}) = \min_{\alpha_i > 0} \frac{n}{\alpha_i} w(\mathfrak{b}_{i,F_\bullet}),$$

we have $w(\sigma) \geq \alpha_i a(w, X, 0)$ for any i and any non-zero $\sigma \in F_i$ if and only if

$$w(\mathfrak{a}_{\alpha,F_\bullet}) \geq n a(w, X, 0),$$

that is, if and only if

$$\operatorname{mult}_E \mathfrak{a}_{\alpha,F_\bullet} > 0 \text{ and } \frac{a(E, X, 0)}{\operatorname{mult}_E \mathfrak{a}_{\alpha,F_\bullet}} \leq \frac{1}{n}. \tag{4.1}$$

For a non-zero ideal $\mathfrak{a}$ on X, a closed point $z \in X$, and a log resolution $g : Y \to X$ of $(X, \mathfrak{a})$, by [OAI26, (2.4)], we have

$$\mathrm{lct}_z(\mathfrak{a}) = \min\left\{ \frac{a(E, X, 0)}{\mathrm{mult}_E \mathfrak{a}} \;\middle|\; E \text{ is a prime divisor on } Y,\ z \in g(E),\ \mathrm{mult}_E \mathfrak{a} > 0 \right\}.$$

Let $z \in X$ be a closed point. If $(z, F_\bullet) \in \Lambda_\alpha$ and w is a divisorial valuation over X with prime divisor E with $z \in \mathrm{center}_X w$ and (4.1), then we take a log resolution of $(X, \mathfrak{a}_{\alpha,F_\bullet})$ on which E is a divisor (such a log resolution exists by [KM98, Theorem 0.2], applied on a smooth projective higher model of X on which E is a divisor), and the formula for lct_z implies that $\mathrm{lct}_z(\mathfrak{a}_{\alpha,F_\bullet}) \le 1/n$. Conversely, if $\mathrm{lct}_z(\mathfrak{a}_{\alpha,F_\bullet}) \le 1/n$, then a prime divisor E which computes the minimum satisfies (4.1), and the divisorial valuation w over X with prime divisor E and constant 1 satisfies $z \in \mathrm{center}_X w$, and thus $(z, F_\bullet) \in \Lambda_\alpha$. Therefore,

$$\Lambda_\alpha = \left\{ (z, F_\bullet) \;\middle|\; \mathrm{lct}_z(\mathfrak{a}_{\alpha,F_\bullet}) \le \frac{1}{n} \right\}.$$

The functions ω/θ are local generators of the ideals $\mathfrak{b}_i$, and thus products of the functions ω/θ are local generators of $\mathfrak{a}_\alpha$ whose restrictions to $X \times \{F_\bullet\}$ generate $\mathfrak{a}_{\alpha,F_\bullet}$. By [OAI26, Lemma 2.3], which does not use flatness of the family of ideals, Λ_α is a Zariski closed subset of $X \times \mathcal{F}$. This proves (1) and (2).

(3) There are only countably many decreasing lists. By definition, U_m° is the complement in U of the union of the $Z_{m,\alpha} \ne X$, and any such $Z_{m,\alpha}$ is a proper Zariski closed subset of X by (1). If a closed point of U_m° realizes α in degree m, then it belongs to $Z_{m,\alpha}$, and hence $Z_{m,\alpha} = X$. □

In Lemma 4.4, different decreasing lists may be realized by different bases and valuations, while the existence of minimizers (Lemma 4.7(3)) and the strict inequality in Section 6 depend on one basis and one valuation, because a sequence of approximations only implies non-strict inequalities. We first fix the basis by the compactness of the complete flag variety, and we then use the compactness of the weights on a log resolution for that basis. The order cannot be reversed because the log resolution depends on the basis (see the first paragraph of [OAI26, Section 4]).

Lemma 4.5 (cf. [OAI26, Proposition 4.4])**.** *Notation and conditions as in Set-up 3.1. Let m be a positive integer with $N_m > 0$, $z \in X$ a closed point, and $p_1, \dots, p_{N_m} \ge 0$ real numbers, not all zero.*

(1) Let s be an ordered basis of R_m, $h : Y \to X$ a log resolution of $(X, \sum_i \mathrm{div}(s_i))$, and $\sum_j E_j$ the reduced divisor on Y with support $\mathrm{Exc}(h) \cup \mathrm{Supp}\, h^* \sum_i \mathrm{div}(s_i)$. Assume that for any real number $\epsilon > 0$, there exists a divisorial valuation w over X with $z \in \mathrm{center}_X w$ and $w(s_i) \ge (p_i - \epsilon) a(w, X, 0)$ for any i. Then there exists a monomial valuation w' on $(Y, \sum_j E_j)$ with $z \in \mathrm{center}_X w'$, $a(w', X, 0) = 1$, and $w'(s_i) \ge p_i$ for any i.

(2) Assume that $p_1 \ge \cdots \ge p_{N_m}$ and that z realizes in degree m any decreasing list α with $\alpha_i < p_i$ for any i with $p_i > 0$ and $\alpha_i = 0$ for any i with $p_i = 0$. Then there exists an ordered basis s of R_m which satisfies the assumption of (1).

Proof. (1) Let $D := \sum_i \mathrm{div}(s_i)$, let i_0 be an index with $p_{i_0} = \max_i p_i > 0$, and let $(\epsilon_k)_k$ be a decreasing sequence of positive real numbers with limit 0 and $\epsilon_1 < p_{i_0}$. For any k, let w_k be a divisorial valuation over X with $z \in \mathrm{center}_X w_k$ and $w_k(s_i) \ge (p_i - \epsilon_k) a(w_k, X, 0)$ for any i. Since $w_k(s_{i_0}) > 0$, we have $w_k(D) > 0$, and hence $u_{k,j} := w_k(E_j) > 0$ for some j. By Lemma 2.2(3), the monomial valuation w_k' on $(Y, \sum_j E_j)$ with weights $u_{k,j}$ along the E_j with $u_{k,j} > 0$, at the irreducible component W_k of $\bigcap_{u_{k,j}>0} E_j$ that contains the center of w_k on Y, satisfies $w_k'(s_i) = w_k(s_i)$ for any i, $a(w_k', X, 0) \le a(w_k, X, 0)$, and $z \in \mathrm{center}_X w_k \subset \mathrm{center}_X w_k'$. Since $w_k'(s_i) \ge 0$, we have

$$w_k'(s_i) \ge (p_i - \epsilon_k) a(w_k', X, 0)$$

for any i. Let $w''_k := w'_k/a(w'_k, X, 0)$, and let $u''_{k,j}$ be the weight of w''_k along E_j, with $u''_{k,j} = 0$ when $u_{k,j} = 0$. Then $w''_k(s_i) \geq p_i - \epsilon_k$ for any i, and $a(w''_k, X, 0) = 1$.

Since X is smooth, the numbers $a(E_j, X, 0)$ are positive integers. Thus $(u''_{k,j})_j$ belongs to the compact simplex

$$\left\{ (u_j)_j \;\middle|\; u_j \geq 0,\ \sum_j a(E_j, X, 0)u_j = 1 \right\}.$$

Since $\sum_j E_j$ has finitely many strata, possibly passing to a subsequence, we may assume that W_k and the set $J := \{j \mid u_{k,j} > 0\}$ do not depend on k, and that $u_j := \lim_{k\to+\infty} u''_{k,j}$ exists for any j. Let $W := W_k$ and $J' := \{j \in J \mid u_j > 0\}$. Then $\sum_{j\in J'} a(E_j, X, 0)u_j = 1$, and in particular, $J' \neq \emptyset$. Let W' be the irreducible component of $\bigcap_{j\in J'} E_j$ which contains W, and let w' be the monomial valuation on $(Y, \sum_j E_j)$ with weights u_j along the E_j with $j \in J'$, at W'. Then $a(w', X, 0) = 1$ and

$$z \in \operatorname{center}_X w'_k = h(W) \subset h(W') = \operatorname{center}_X w'.$$

Let $c_{ij} := \operatorname{mult}_{E_j} h^* \operatorname{div}(s_i)$. Since $\sum_j E_j$ is SNC and $\operatorname{Supp} h^* \operatorname{div}(s_i) \subset \operatorname{Supp} \sum_j E_j$, a local equation of $h^* \operatorname{div}(s_i)$ at the generic point of W is the product of a unit and a monomial in local equations of the E_j with $j \in J$, with exponents c_{ij}, and a local equation of $h^* \operatorname{div}(s_i)$ at the generic point of W' is the product of a unit and a monomial in local equations of the E_j with $j \in J'$, with exponents c_{ij}. Thus

$$w''_k(s_i) = \sum_{j\in J} u''_{k,j}c_{ij} \text{ and } w'(s_i) = \sum_{j\in J'} u_j c_{ij} = \lim_{k\to+\infty} w''_k(s_i) \geq p_i$$

for any i.

(2) For any positive integer k and any i, let

$$\alpha_i^{(k)} := \frac{\lfloor k!(1 - 1/k)p_i \rfloor}{k!}.$$

Then $\alpha^{(k)}$ is a decreasing list, and $0 \leq \alpha_i^{(k)} \leq (1 - 1/k)p_i$ for any i. Since multiples of $1/k!$ are multiples of $1/(k + 1)!$ and $1 - 1/k \leq 1 - 1/(k + 1)$, we have $\alpha_i^{(k)} \leq \alpha_i^{(k+1)}$ for any i. Moreover, $\lim_{k\to+\infty} \alpha_i^{(k)} = p_i$ for any i. By assumption, z realizes $\alpha^{(k)}$ in degree m for any k.

For any k, let Λ_k be the set of complete flags $F_1 \subset \cdots \subset F_{N_m} = R_m$ with $\dim F_i = i$ for which there exists a divisorial valuation w over X with $z \in \operatorname{center}_X w$ and $w(\sigma) \geq \alpha_i^{(k)} a(w, X, 0)$ for any i and any non-zero $\sigma \in F_i$. By Lemma 4.4(2), Λ_k is Zariski closed in the complete flag variety of R_m. Since z realizes $\alpha^{(k)}$ in degree m, there exist an ordered basis $s^{(k)}$ of R_m and a divisorial valuation w over X with $z \in \operatorname{center}_X w$ and $w(s_i^{(k)}) \geq \alpha_i^{(k)} a(w, X, 0)$ for any i. Since $\alpha^{(k)}$ is decreasing, by [JM12, Proposition 3.1], the flag spanned by $s^{(k)}$ belongs to Λ_k, and thus $\Lambda_k \neq \emptyset$. Since $\alpha_i^{(k)} \leq \alpha_i^{(k+1)}$ for any i, we have $\Lambda_{k+1} \subset \Lambda_k$. Since the complete flag variety of R_m is a Noetherian topological space, $\bigcap_k \Lambda_k \neq \emptyset$.

Let $F_1 \subset \cdots \subset F_{N_m} = R_m$ be a flag in $\bigcap_k \Lambda_k$, and let s be an ordered basis of R_m such that $s_1, \ldots, s_i$ span F_i for any i. For any real number $\epsilon > 0$, there exists k such that $\alpha_i^{(k)} \geq p_i - \epsilon$ for any i. Since $s_i \in F_i$, a divisorial valuation w over X with $z \in \operatorname{center}_X w$ and $w(\sigma) \geq \alpha_i^{(k)} a(w, X, 0)$ for any i and any non-zero $\sigma \in F_i$ satisfies $w(s_i) \geq (p_i - \epsilon)a(w, X, 0)$ for any i. Thus s satisfies the assumption of (1). □

Lemma 4.6 (cf. [OAI26, Lemma 4.5])**.** *Notation and conditions as in Set-up 3.1. Let m be a positive integer with $N_m > 0$, t a positive real number, V a closed subvariety of X, and $x \in V$ a closed point. Then $\mu_{m,t}(y) \geq \mu_{m,t}(x)$ for any very general closed point y of V.*

Proof. By Lemma 4.4(1), for any decreasing list α, either $Z_{m,\alpha} \cap V = V$ or $Z_{m,\alpha} \cap V$ is a proper Zariski closed subset of V. Since there are only countably many decreasing lists, a very general

closed point y of V does not belong to any proper subset $Z_{m,\alpha} \cap V$ of V. Thus any decreasing list realized in degree m by y is realized in degree m by x.

By Lemma 2.2(1), it suffices to prove that $\psi_{m,t}(w,s) \geq \mu_{m,t}(x)$ for any divisorial valuation w over X with $y \in \operatorname{center}_X w$ and any ordered basis s of R_m. Let $a := a(w,X,0)$. Since $\psi_{m,t}(w,s)$ does not depend on the order of s, possibly reordering s, we may assume that the numbers $p_i := w(s_i)/a$ satisfy $p_1 \geq \cdots \geq p_{N_m}$.

Let k be a positive integer, and let $\alpha_i := \lfloor kp_i \rfloor / k$ for any i. Then α is a decreasing list with $p_i - 1/k < \alpha_i \leq p_i$ for any i, and y realizes α in degree m. Thus there exist an ordered basis s' of R_m and a divisorial valuation w' over X with $x \in \operatorname{center}_X w'$ and $w'(s'_i) \geq \alpha_i a(w',X,0)$ for any i. Let $w'' := (a/a(w',X,0))w'$. Then w'' is a divisorial valuation over X with $x \in \operatorname{center}_X w''$ and $a(w'',X,0) = a$, and

$$w''(s'_i) \geq \alpha_i a \geq w(s_i) - \frac{a}{k}$$

for any i. Therefore,

$$\mu_{m,t}(x) \leq \psi_{m,t}(w'',s') \leq \frac{1}{N_m} \sum_{i=1}^{N_m} \exp\left(\frac{a}{t} - \frac{w(s_i)}{m} + \frac{a}{km}\right) = \exp\left(\frac{a}{km}\right) \psi_{m,t}(w,s).$$

Letting $k \to +\infty$, we obtain $\mu_{m,t}(x) \leq \psi_{m,t}(w,s)$. □

4.3. **Existence of minimizers.** In this subsection, we prove that $\psi_{m,t}$ has a minimizer at any closed point of U, with bounds on $\mu_{m,t}$ and on the log discrepancy that are uniform in m and t. The upper bound of Lemma 4.7(1) follows from the condition $t_-(L^d)^{1/d} > d+1$ and a lattice point count (see (b) in the sketch of the proof in Section 1). We work on an interval $[t_-, t_+]$ because in Section 5, when $L^d \geq 2$, the number t is fixed only after the lower bound on the log discrepancy is known, and thus the constants of Lemma 4.7 must not depend on t.

Lemma 4.7 (cf. [OAI26, Lemma 4.2 and Proposition 4.4])**.** *Notation and conditions as in Set-up 3.1. Let $x \in U$ be a closed point and $t_- \leq t_+$ positive real numbers. Assume that $t_-(L^d)^{1/d} > d+1$. Then there exist real numbers $\delta \in (0,1)$, $\kappa > 0$, $a_* > 0$, $a^* > 0$, and a positive integer m_0 such that for any integer $m \geq m_0$ and any real number $t \in [t_-, t_+]$, the following hold.*

1. *$\kappa \leq \mu_{m,t}(x) \leq 1 - \delta$.*
2. *For any monomial valuation v over X with $x \in \operatorname{center}_X v$ and any ordered basis s of R_m with $\psi_{m,t}(v,s) \leq 1 - \delta/2$, we have $a_* \leq a(v,X,0) \leq a^*$.*
3. *$\psi_{m,t}$ has a minimizer at x.*

Proof. The proof is long, so we divide it into several steps. Since L is big, there exists a positive integer m_0 with $N_m > 0$ for any integer $m \geq m_0$. We enlarge m_0 finitely many times below.

Step 1. In this step we prove that there exists a real number $C > 0$ with

$$\operatorname{mult}_x \operatorname{div}(\sigma) \leq Cm \tag{4.2}$$

for any positive integer m and any non-zero $\sigma \in R_m$.

Let A be a very ample divisor on X, $C := (L \cdot A^{d-1})$, $h : Y \to X$ the blow-up of X at x, and E the exceptional divisor. Since A is very ample, $|h^*A - E|$ is base point free, hence $h^*A - E$ is nef. Let m be a positive integer, $\sigma \in R_m$ a non-zero section, and $D := \operatorname{div}(\sigma)$. Then $D' := h^*D - (\operatorname{mult}_x D)E \geq 0$. We have $((h^*A - E)^{d-1} \cdot h^*D) = (A^{d-1} \cdot D)$ by the projection formula, and $((h^*A - E)^{d-1} \cdot E) = 1$ because $-E|_E$ is $\mathcal{O}_{\mathbb{P}^{d-1}}(1)$ on $E \cong \mathbb{P}^{d-1}$. Thus [Laz04, Theorem 1.4.9] implies that

$$0 \leq ((h^*A - E)^{d-1} \cdot D') = m(L \cdot A^{d-1}) - \operatorname{mult}_x D,$$

which is (4.2). Since L is big, $N_m \geq 2$ for any sufficiently large m, so R_m contains a non-zero section vanishing at x, and (4.2) implies that $C > 0$.

Step 2. In this step we prove (4.5) below by a lattice point count.

Fix regular parameters at x and a local generator of $\mathcal{O}_X(L)$ at x. Order the monomials in the regular parameters first by total degree and then lexicographically. For a non-zero $\sigma \in R_m$, we let $\gamma(\sigma) \in \mathbb{Z}^d_{\geq 0}$ be the exponent of the first non-zero Taylor coefficient of σ at x, computed with the m-th power of the local generator, where the germ of σ at x is non-zero because X is integral. The multiplicity of $\operatorname{div}(\sigma)$ at x is $|\gamma(\sigma)|$. Let $H_m := \{\gamma(\sigma) \mid 0 \neq \sigma \in R_m\}$. By Gaussian elimination, $\#H_m = N_m$ and R_m has an ordered basis $(s_1, \ldots, s_{N_m})$ with $H_m = \{\gamma(s_1), \ldots, \gamma(s_{N_m})\}$.

Since L is nef and big, $L^d > 0$ by [Laz04, Theorem 2.2.16], and

$$N_m = \frac{L^d}{d!} m^d + O(m^{d-1})$$

by [Laz04, Corollary 1.4.41]. For $m \geq m_0$, let l_m be the largest integer with $\binom{l_m+d}{d} \leq N_m$. Then $N_m < \binom{l_m+d+1}{d}$, and the asymptotic formula for N_m implies that $\lim_{m\to+\infty} l_m/m = (L^d)^{1/d}$. A direct calculation shows that for any integer $l \geq 0$,

$$\sum_{\gamma \in \mathbb{Z}^d_{\geq 0}, |\gamma| \leq l} |\gamma| = d\binom{l+d}{d+1} = \frac{dl}{d+1}\binom{l+d}{d}. \tag{4.3}$$

Among all sets of N_m distinct points of $\mathbb{Z}^d_{\geq 0}$, the sum of the total degrees is smallest when the points of total degree at most l_m are all taken and the other $N_m - \binom{l_m+d}{d}$ points have total degree $l_m + 1$. Since $l_m + 1 \geq \frac{dl_m}{d+1}$, (4.3) implies that

$$\frac{1}{N_m} \sum_{\gamma \in H_m} |\gamma| \geq \frac{dl_m}{d+1}. \tag{4.4}$$

Dividing by m and using $\lim_{m\to+\infty} l_m/m = (L^d)^{1/d}$, we obtain (cf. [OAI26, (3.3)])

$$\liminf_{m\to+\infty} \frac{1}{mN_m} \sum_{\gamma \in H_m} |\gamma| \geq (L^d)^{1/d} \frac{d}{d+1} > \frac{d}{t_-},$$

where the last inequality is the assumption $t_-(L^d)^{1/d} > d + 1$. Thus there exists a real number $\epsilon > 0$ such that, possibly enlarging m_0, we have

$$\frac{1}{mN_m} \sum_{\gamma \in H_m} |\gamma| \geq \frac{d}{t_-} + 3\epsilon \geq \frac{d}{t} + 3\epsilon \tag{4.5}$$

for any $m \geq m_0$ and any $t \in [t_-, t_+]$.

Step 3. In this step we prove the upper bound of (1).

Recall that $h : Y \to X$ is the blow-up of X at x and E is the exceptional divisor. For a real number $\lambda \in (0, 1]$, let v_λ be the divisorial valuation over X with prime divisor E and constant $c = \lambda$ in Definition 2.1. Then $\operatorname{center}_X v_\lambda = \{x\}$, $a(v_\lambda, X, 0) = \lambda d$, and $v_\lambda(\sigma) = \lambda \operatorname{mult}_x \operatorname{div}(\sigma)$ for any non-zero $\sigma \in R_m$.

Let $m \geq m_0$ be an integer, $t \in [t_-, t_+]$, $s = (s_1, \ldots, s_{N_m})$ the ordered basis of **Step 2**, and

$$y_i := \lambda\left(\frac{d}{t} - \frac{\operatorname{mult}_x \operatorname{div}(s_i)}{m}\right), \qquad C_1 := \frac{d}{t_-} + C.$$

Then $\psi_{m,t}(v_\lambda, s) = \frac{1}{N_m}\sum_i e^{y_i}$, and (4.2) implies that $|y_i| \leq \lambda C_1$. By (4.5), $\frac{1}{N_m}\sum_i y_i \leq -3\epsilon\lambda$. Since $e^y \leq 1 + y + \frac{1}{2}y^2 e^{|y|}$ for any real number y, we have

$$\psi_{m,t}(v_\lambda, s) \leq 1 - 3\epsilon\lambda + \frac{1}{2}\lambda^2 C_1^2 e^{C_1}. \tag{4.6}$$

Fix $\lambda \in (0, 1]$ with $\frac{1}{2}\lambda C_1^2 e^{C_1} \leq \epsilon$, and let $\delta := \min\{2\epsilon\lambda, 1/2\}$. The numbers λ and δ do not depend on m or t, and (4.6) implies that

$$\mu_{m,t}(x) \leq \psi_{m,t}(v_\lambda, s) \leq 1 - 2\epsilon\lambda \leq 1 - \delta.$$

Step 4. In this step we prove the lower bound of (1) and prove (2).

By Proposition 3.2(2) with $V = X$ and $y = x$, there exist a positive integer k and $\tau_0, \dots, \tau_d \in R_k$ with $\tau_0(x) \neq 0$ and $\tau_i(x) = 0$ for any $i \geq 1$ such that $z_i := \tau_i/\tau_0$, $1 \leq i \leq d$, form a regular system of parameters of $\mathcal{O}_{X,x}$. By Lemma 2.4(3), there exists a positive integer k' such that $|lL|$ has no base point in U for any integer $l \geq k'$. For any integer $0 \leq c < k$, we choose an integer $b_c \geq k'$ with $b_c \equiv c \pmod{k}$ and a section σ_c of $b_c L$ with $\sigma_c(x) \neq 0$. Possibly enlarging m_0, we may assume that $m_0 \geq b_c$ for any c.

Let $m \geq m_0$, write $m = q'k + b_c$ with integers $q' \geq 0$ and $0 \leq c < k$, and let

$$r_m := \min\left\{q', \left\lfloor \frac{m}{2t_+} \right\rfloor\right\}.$$

For $\gamma \in \mathbb{Z}^d_{\geq 0}$ with $|\gamma| \leq r_m$, we let

$$\sigma_\gamma := \sigma_c \tau_0^{q'-|\gamma|} \prod_{i=1}^{d} \tau_i^{\gamma_i} \in R_m.$$

Near x, σ_γ is $\sigma_c \tau_0^{q'}$ times z^γ, and $\sigma_c \tau_0^{q'}$ does not vanish at x. Let $t \in [t_-, t_+]$, and let $R'_{m,t}$ be the linear subspace of R_m consisting of 0 and of the non-zero σ with $\operatorname{mult}_x \operatorname{div}(\sigma) > m/(2t)$. Since $r_m \leq m/(2t_+) \leq m/(2t)$, any non-trivial linear combination of the σ_γ has multiplicity at most $m/(2t)$ at x. Thus the σ_γ are linearly independent modulo $R'_{m,t}$, and

$$\operatorname{codim}_{R_m} R'_{m,t} \geq \binom{r_m + d}{d}.$$

Since $\liminf_{m\to+\infty} r_m/m > 0$ (as q'/m tends to $1/k$) and $N_m = O(m^d)$, possibly enlarging m_0, there exists a real number $\kappa > 0$, which does not depend on m or t, with

$$\operatorname{codim}_{R_m} R'_{m,t} \geq \kappa N_m \tag{4.7}$$

for any $m \geq m_0$ and any $t \in [t_-, t_+]$.

Let v be a monomial valuation over X with $x \in \operatorname{center}_X v$ and $s = (s_1, \dots, s_{N_m})$ an ordered basis of R_m. By Lemma 2.2(2), for any non-zero $\sigma \in R_m$,

$$v(\sigma) \leq a(v, X, 0) \operatorname{mult}_x \operatorname{div}(\sigma). \tag{4.8}$$

Thus any s_i with $v(s_i) > ma(v, X, 0)/(2t)$ belongs to $R'_{m,t}$, and (4.7) implies that at least κN_m indices i satisfy $v(s_i) \leq ma(v, X, 0)/(2t)$. For such an i, we have

$$\exp\left(\frac{a(v, X, 0)}{t} - \frac{v(s_i)}{m}\right) \geq \exp\left(\frac{a(v, X, 0)}{2t}\right).$$

Since $a(v, X, 0) > 0$ and $t \leq t_+$, we obtain

$$\psi_{m,t}(v, s) \geq \kappa \exp\left(\frac{a(v, X, 0)}{2t_+}\right) \geq \kappa. \tag{4.9}$$

Thus $\mu_{m,t}(x) \geq \kappa$, which proves (1). In particular, $\kappa \leq 1 - \delta < 1$. Moreover, (4.8) and (4.2) imply that $v(s_i)/m \leq Ca(v, X, 0)$ for any i, hence

$$\psi_{m,t}(v, s) \geq \exp\left(\frac{a(v, X, 0)}{t} - Ca(v, X, 0)\right) \geq \exp(-Ca(v, X, 0)). \tag{4.10}$$

Let

$$a^* := 2t_+ \log\frac{1}{\kappa}, \qquad a_* := \frac{1}{C} \log\frac{1}{1 - \delta/2}.$$

If $\psi_{m,t}(v, s) \leq 1 - \delta/2$, then (4.9) and (4.10) imply that $a_* \leq a(v, X, 0) \leq a^*$, which proves (2).

Step 5. In this step we prove (3).

Fix $m \geq m_0$ and $t \in [t_-, t_+]$. For a monomial valuation v and an ordered basis s of R_m, Lemma 2.2(1) with $\mathcal{S} = \{s_1, \dots, s_{N_m}\}$ implies that $\psi_{m,t}(v, s)$ is a limit of values $\psi_{m,t}(v', s)$ with

v' divisorial and $\mathrm{center}_X v' = \mathrm{center}_X v$. Thus there exist divisorial valuations v_n over X with $x \in \mathrm{center}_X v_n$ and ordered bases $s^{(n)}$ of R_m with

$$\lim_{n\to+\infty} \psi_{m,t}(v_n, s^{(n)}) = \mu_{m,t}(x).$$

By (1), we may assume that $\psi_{m,t}(v_n, s^{(n)}) \le 1 - \delta/2$ for any n, hence $a(v_n, X, 0) \in [a_*, a^*]$ by (2). Possibly reordering each $s^{(n)}$, we may assume that the numbers $v_n(s_i^{(n)})/a(v_n, X, 0)$ decrease with i, and they belong to $[0, Cm]$ by (4.8) and (4.2). Possibly passing to a subsequence, we may assume that

$$a_0 := \lim_{n\to+\infty} a(v_n, X, 0) \in [a_*, a^*], \qquad p_i := \lim_{n\to+\infty} \frac{v_n(s_i^{(n)})}{a(v_n, X, 0)}$$

exist, where $p_1 \ge \cdots \ge p_{N_m} \ge 0$. Then

$$\mu_{m,t}(x) = \frac{1}{N_m} \sum_{i=1}^{N_m} \exp\left(\frac{a_0}{t} - \frac{a_0 p_i}{m}\right). \tag{4.11}$$

Not all p_i are zero, since otherwise (4.11) implies that $\mu_{m,t}(x) = \exp(a_0/t) > 1$, which contradicts (1).

Let α be a decreasing list with $\alpha_i < p_i$ for any i with $p_i > 0$ and $\alpha_i = 0$ for any i with $p_i = 0$. For any sufficiently large n, we have $v_n(s_i^{(n)}) \ge \alpha_i a(v_n, X, 0)$ for any i, so x realizes α in degree m. By Lemma 4.5, there exist an ordered basis s' of R_m and a monomial valuation w' over X with $x \in \mathrm{center}_X w'$, $a(w', X, 0) = 1$, and $w'(s'_i) \ge p_i$ for any i. By Definition 2.1, $x \in \mathrm{center}_X(a_0 w')$, $a(a_0 w', X, 0) = a_0$, and $(a_0 w')(s'_i) \ge a_0 p_i$ for any i. Thus (4.11) implies that

$$\mu_{m,t}(x) \le \psi_{m,t}(a_0 w', s') \le \frac{1}{N_m} \sum_{i=1}^{N_m} \exp\left(\frac{a_0}{t} - \frac{a_0 p_i}{m}\right) = \mu_{m,t}(x).$$

Therefore $(a_0 w', s')$ is a minimizer of $\psi_{m,t}$ at x. □

5. Minimizers centered at a point

The goal of this section is to prove the following proposition.

Proposition 5.1 (cf. [OAI26, Proposition 5.1])**.** *Notation and conditions as in Set-up 3.1. Let $x \in U$ be a closed point. Then:*

1. *For any real number $t > d+1$, there exist infinitely many positive integers m such that (m, t) is good at x.*
2. *If $L^d \ge 2$, then there exist a real number t with $0 < t < d+1$ and infinitely many positive integers m such that (m, t) is good at x.*

Section 6 needs minimizers whose center is a point. If the center is positive-dimensional, then we obtain two bounds on one quantity, an upper bound from the first variation in the tangential directions (Subsection 5.1) and a lower bound from the growth of slices (Subsection 5.2), and the upper bound and the lower bound are incompatible as $m \to +\infty$. We prove Proposition 5.1 at the end of Subsection 5.2.

Since [OAI26, Proposition 5.1] assumes that L is ample and that $K_X + (d+1)L$ is not generated at x, we write out the proof, which uses the method of [OAI26, Section 5].

5.1. Taylor expansions along the center.

In this subsection, we expand sections in local parameters at a very general point of the center of a minimizer, and we prove the upper bound from the first variation in the tangential directions.

Construction-Proposition 5.2. Notation and conditions as in Set-up 3.1. Let k be a positive integer which satisfies Proposition 3.2(2), m a positive integer with $N_m > 0$, t a positive real number, $x \in U$ a closed point, and (v, s) a minimizer of $\psi_{m,t}$ at x such that $V := \mathrm{center}_X v$ is

not a point. Let $e := \dim V$. Write v as the monomial valuation over X with weights $u_1, \dots, u_r$ along $E_1, \dots, E_r$ at W, with $h : Y \to X$, Y^0, and $\sum_j E_j$ from Definition 2.1.

(1) Let $y \in V \cap U$ be a smooth point of V with $\mu_{m,t}(y) \geq \mu_{m,t}(x)$, and $y' \in W$ a closed point with $h(y') = y$ such that y' is not on any component of $\sum_j E_j$ other than $E_1, \dots, E_r$ and $h|_W : W \to V$ is smooth at y'.
(2) Let $\tau_0, \dots, \tau_e \in R_k$ be sections for V and y from Proposition 3.2(2), and $\xi_i := h^*(\tau_i/\tau_0)$ for $1 \leq i \leq e$.
(3) Let $\zeta_1, \dots, \zeta_r$ be local equations of $E_1, \dots, E_r$ at y', and $\eta_1, \dots, \eta_{d-e-r}$ regular functions at y' such that $\zeta_1, \dots, \zeta_r$, $\eta_1, \dots, \eta_{d-e-r}$, and $\xi_1, \dots, \xi_e$ form a regular system of parameters of $\mathcal{O}_{Y,y'}$.
(4) For a positive integer j and a non-zero $\sigma \in R_j$, expand $h^*\sigma$ as a power series in the parameters of (3), in the trivialization of $h^*\mathcal{O}_X(jL)$ near y' by the j-th power of a fixed local generator of $\mathcal{O}_X(L)$ at y, and write the exponents as (γ, β) with $\gamma \in \mathbb{Z}^{d-e}_{\geq 0}$ and $\beta \in \mathbb{Z}^e_{\geq 0}$, where γ_i is the exponent of ζ_i for $i \leq r$ and of η_{i-r} for $i > r$, and β_i is the exponent of ξ_i.
(5) With $\ell(\gamma) := \sum_{i=1}^r u_i\gamma_i$ and $|\beta| := \sum_{i=1}^e \beta_i$, order the exponents first by $\ell(\gamma)$, then by $|\beta|$, and then by the graded lexicographic order.
(6) The *leading exponent* of σ is the least exponent with a non-zero coefficient in the expansion of $h^*\sigma$, and Γ_j is the set of leading exponents of non-zero elements of R_j.
(7) For a positive integer j with $N_j > 0$, we let

$$Q_j := \sum_{(\gamma,\beta)\in\Gamma_j} \exp\left(-\frac{\ell(\gamma)}{m}\right), \quad T_m := \frac{1}{mQ_m} \sum_{(\gamma,\beta)\in\Gamma_m} |\beta| \exp\left(-\frac{\ell(\gamma)}{m}\right).$$

(8) For $\gamma \in \mathbb{Z}^{d-e}_{\geq 0}$, we call $G_\gamma(j) := \{\beta \mid (\gamma, \beta) \in \Gamma_j\}$ a *slice* of Γ_j.
(9) For $\gamma \in \mathbb{Z}^{d-e}_{\geq 0}$, we let $b_\gamma(j) := (e!\#G_\gamma(j))^{1/e}$.

Then $e \leq d-1$, the points y and y' of (1) and the functions $\zeta_1, \dots, \zeta_r$, $\eta_1, \dots, \eta_{d-e-r}$ of (3) exist, and the order in (5) is a well-order compatible with addition.

Proof. Since (v, s) is a minimizer of $\psi_{m,t}$ at x, we have $x \in V \cap U$. Since $r \geq 1$ and V is contained in the closure of $h(E_1)$, we have $e \leq \dim E_1 = d - 1$. In particular, Proposition 3.2(1) can be used for V.

We construct y and y'. Let W_0 be the complement in W of the other components of $\sum_j E_j$. Since $\sum_j E_j$ is SNC, W_0 is a smooth variety, and W_0 dominates V. By Chevalley's theorem [Har77, Exercise II.3.19], $h(W_0)$ contains a dense open subset of V. Thus, by generic smoothness [Har77, Corollary III.10.7], there exists a dense open subset V_0 of V which is contained in $h(W_0)$ and in the smooth locus of V, such that $W_0 \cap h^{-1}(V_0) \to V_0$ is smooth. Since $x \in V \cap U$ and V is irreducible, $V_0 \cap U$ is a non-empty open subset of V. Since $\mathbb{C}$ is uncountable, a very general closed point of V belongs to $V_0 \cap U$. Thus, by Lemma 4.6, we may pick a closed point $y \in V_0 \cap U$ with $\mu_{m,t}(y) \geq \mu_{m,t}(x)$. Since $y \in h(W_0)$, we may pick a closed point $y' \in W_0$ with $h(y') = y$. Then y and y' satisfy (1).

We construct the parameters of (3). Since $h|_W$ is smooth at y', the map from the tangent space of W at y' to the tangent space of V at y is surjective [Har77, Proposition III.10.4], so the map between the cotangent spaces in the other direction is injective. Since $(\tau_1/\tau_0)|_V, \dots, (\tau_e/\tau_0)|_V$ form a regular system of parameters of $\mathcal{O}_{V,y}$, the functions $\xi_1|_W, \dots, \xi_e|_W$ are linearly independent in the cotangent space of W at y'. Since $\dim W = d - r \geq e$, there exist $\bar{\eta}_1, \dots, \bar{\eta}_{d-e-r} \in \mathcal{O}_{W,y'}$ which form a regular system of parameters of $\mathcal{O}_{W,y'}$ together with $\xi_1|_W, \dots, \xi_e|_W$. We let $\eta_1, \dots, \eta_{d-e-r}$ be regular functions at y' with $\eta_i|_W = \bar{\eta}_i$. Since $\zeta_1, \dots, \zeta_r$ generate the ideal of W in $\mathcal{O}_{Y,y'}$, (3) holds.

For any real number c, only finitely many $(\gamma_1, \dots, \gamma_r)$ satisfy $\sum_{i=1}^r u_i\gamma_i \leq c$, so the set of values of ℓ has no infinite decreasing sequence. Since ℓ and $|\beta|$ are additive, and the graded

lexicographic order is a well-order compatible with addition, the order in (5) is a well-order compatible with addition. □

The first two levels of the order in Construction-Proposition 5.2(5) compute v and the first variation in the tangential directions in Lemma 5.4. Here kL need not be very ample, so the sections $\tau_0, \dots, \tau_e$ come from Proposition 3.2(2) in place of the embedding in [OAI26, Section 5.2], and Lemma 5.5 depends on the fact that k does not depend on V. The denominator in the exponent of Q_j is the fixed integer m, so the weight $\exp(-\ell(\gamma)/m)$ does not depend on j, and Lemma 5.5 can compare slices of different degrees under one weight.

Lemma 5.3. *Notation as in Construction-Proposition 5.2. Let j and l be positive integers with $N_j > 0$ and $N_l > 0$. Then:*

(1) $\#\Gamma_j = N_j$ and $\Gamma_j + \Gamma_l \subset \Gamma_{j+l}$.
(2) $v(\sigma) = \ell(\gamma)$ for any non-zero $\sigma \in R_j$ with leading exponent (γ, β).
(3) $Q_m = N_m \mu_{m,t}(x) \exp(-a(v, X, 0)/t)$ and $Q_j \geq N_j \mu_{j,t}(x) \exp(-ja(v, X, 0)/(mt))$.
(4) $\#G_0(j) \geq I_V(j)$ and $\{0, 1\}^e \subset G_0(ke)$.

Proof. (1) Since Y is integral and h is birational, the expansion of $h^*\sigma$ is non-zero for any non-zero $\sigma \in R_j$. If the leading exponents of two non-zero elements of R_j are equal, then a linear combination of them is zero or has a larger leading exponent. Thus Gaussian elimination on the coefficients of the least exponents implies that $\#\Gamma_j = \dim R_j = N_j$, and that R_j has an ordered basis whose leading exponents are exactly Γ_j.

Since the order is compatible with addition, the leading exponent of a product of two non-zero power series is the sum of the leading exponents. Since the local generators in degrees j, l, and $j + l$ are powers of one local generator, the expansion of $h^*(\sigma\sigma')$ is the product of the expansions of $h^*\sigma$ and $h^*\sigma'$ for any non-zero $\sigma \in R_j$ and $\sigma' \in R_l$. Thus $\Gamma_j + \Gamma_l \subset \Gamma_{j+l}$.

(2) Since y' is not on the other components of $\sum_j E_j$, [OAI26, Lemma 2.1] implies that $v(\sigma)$ is the minimum of $\ell(\gamma')$ over the exponents (γ', β') with a non-zero coefficient in the expansion of $h^*\sigma$. Since the order compares ℓ first, the minimum is $\ell(\gamma)$.

(3) By (1), there exists an ordered basis $\bar{s} = (\bar{s}_1, \dots, \bar{s}_{N_m})$ of R_m whose leading exponents are exactly Γ_m. Let $\sigma = \sum_i c_i \bar{s}_i$ be a non-zero element of R_m. Since the leading exponents of the $\bar{s}_i$ are distinct, the leading exponent of σ is the least leading exponent of the $\bar{s}_i$ with $c_i \neq 0$. Thus (2) implies that $v(\sigma) = \min\{v(\bar{s}_i) \mid c_i \neq 0\}$, so $\bar{s}$ is adapted to v. By Lemma 4.2(1), with $\bar{s}$ as the adapted basis and s as the arbitrary basis, we have

$$\psi_{m,t}(v, \bar{s}) \leq \psi_{m,t}(v, s) = \mu_{m,t}(x).$$

Since $x \in V = \operatorname{center}_X v$, we have $\psi_{m,t}(v, \bar{s}) \geq \mu_{m,t}(x)$. By (2), we obtain

$$\mu_{m,t}(x) = \psi_{m,t}(v, \bar{s}) = \frac{1}{N_m} \sum_{(\gamma,\beta)\in\Gamma_m} \exp\left(\frac{a(v, X, 0)}{t} - \frac{\ell(\gamma)}{m}\right) = \frac{Q_m}{N_m} \exp\left(\frac{a(v, X, 0)}{t}\right),$$

which is the first formula of (3).

By Definition 2.1, $\operatorname{center}_X((j/m)v) = V \ni x$ and $a((j/m)v, X, 0) = (j/m)a(v, X, 0)$. By (1), there exists an ordered basis $\bar{s}' = (\bar{s}'_1, \dots, \bar{s}'_{N_j})$ of R_j whose leading exponents are exactly Γ_j. By (2), we have

$$\mu_{j,t}(x) \leq \psi_{j,t}\left(\frac{j}{m}v, \bar{s}'\right) = \frac{1}{N_j} \sum_{(\gamma,\beta)\in\Gamma_j} \exp\left(\frac{ja(v, X, 0)}{mt} - \frac{\ell(\gamma)}{m}\right) = \frac{Q_j}{N_j} \exp\left(\frac{ja(v, X, 0)}{mt}\right),$$

which is the second formula of (3).

(4) Let θ be the fixed local generator of $\mathcal{O}_X(L)$ at y, and $z_i := (\tau_i/\tau_0)|_V$ for $1 \leq i \leq e$. Let $\sigma \in R_j$ with $\sigma|_V \neq 0$, and write $\sigma = f\theta^j$ with $f \in \mathcal{O}_{X,y}$. Since V is integral and $z_1, \dots, z_e$ form a regular system of parameters of $\mathcal{O}_{V,y}$, we may write $f|_V = \sum_\beta c_\beta z^\beta$ in the completion of $\mathcal{O}_{V,y}$,

with some $c_\beta \neq 0$. Near y', W is the zero locus of $\zeta_1, \dots, \zeta_r$, and $\xi_i|_W = (h|_W)^* z_i$. Thus

$$(h^* f)|_W = \sum_\beta c_\beta (\xi|_W)^\beta.$$

In particular, the expansion of $h^*\sigma$ with $\zeta_1 = \cdots = \zeta_r = 0$ contains only monomials in $\xi_1, \dots, \xi_e$, and no η_i. Since $u_i > 0$ for any i, an exponent (γ, β) satisfies $\ell(\gamma) = 0$ if and only if $\gamma_1 = \cdots = \gamma_r = 0$. Thus the exponents (γ, β) with $\ell(\gamma) = 0$ and a non-zero coefficient in the expansion of $h^*\sigma$ are exactly the $(0, \beta)$ with $c_\beta \neq 0$. Therefore, the leading exponent of σ is $(0, \beta_0)$, where β_0 is the least β with $c_\beta \neq 0$ for the order by $|\beta|$ and then by the graded lexicographic order. Gaussian elimination on the image of R_j in $H^0(V, jL|_V)$, as in (1), implies that β_0 takes $I_V(j)$ distinct values. Thus $\#G_0(j) \geq I_V(j)$.

Let $\beta \in \{0,1\}^e$. Then

$$\sigma_\beta := \tau_0^{e-|\beta|} \prod_{i=1}^{e} \tau_i^{\beta_i} \in R_{ke},$$

and the expansion of $h^*\sigma_\beta$ is $h^*(\tau_0^e/\theta^{ke})\xi^\beta$. Since $\tau_0(y) \neq 0$, $h^*(\tau_0^e/\theta^{ke})$ is a unit at y', so the leading exponent of $h^*(\tau_0^e/\theta^{ke})$ is zero. Since the leading exponent of a product of two non-zero power series is the sum of the leading exponents, the leading exponent of σ_β is $(0, \beta)$. Thus $\{0,1\}^e \subset G_0(ke)$. □

The monomial valuation in the proof below perturbs v at y' and has center $\{y\}$, which need not contain x. The inequality in Construction-Proposition 5.2(1) is there for that reason.

Lemma 5.4. *Notation as in Construction-Proposition 5.2. Then $T_m \leq e/t$.*

Proof. By Lemma 5.3(1), there exists an ordered basis $\bar{s} = (\bar{s}_1, \dots, \bar{s}_{N_m})$ of R_m whose leading exponents are exactly Γ_m. Let $(\gamma^{(i)}, \beta^{(i)})$ be the leading exponent of $\bar{s}_i$. By Lemma 5.3(2), $v(\bar{s}_i) = \ell(\gamma^{(i)})$. Thus Lemma 5.3(3) implies that

$$\psi_{m,t}(v, \bar{s}) = \frac{Q_m}{N_m} \exp\left(\frac{a(v, X, 0)}{t}\right) = \mu_{m,t}(x).$$

Since $\zeta_1, \dots, \zeta_r$, $\eta_1, \dots, \eta_{d-e-r}$, and $\xi_1, \dots, \xi_e$ form a regular system of parameters of $\mathcal{O}_{Y,y'}$, there exists an open neighborhood Y^1 of y' in Y^0 which does not intersect the other components of $\sum_j E_j$, such that the zero loci of the ζ_i, the η_i, and the ξ_i in Y^1 are the components of a reduced SNC divisor $\sum_l F_l$ on Y^1 with $\bigcap_l F_l = \{y'\}$. Since $\operatorname{Supp} K_{Y/X}|_{Y^1}$ is contained in $E_1 \cup \cdots \cup E_r$, h, Y^1, and $\sum_l F_l$ are as in Definition 2.1, and the zero loci of the η_i and the ξ_i have log discrepancy 1.

For a real number $\epsilon > 0$, let v_ϵ be the monomial valuation over X at the stratum $\{y'\}$, with weights u_i along $\{\zeta_i = 0\}$, ϵ^2 along $\{\eta_i = 0\}$, and ϵ along $\{\xi_i = 0\}$. Then $\operatorname{center}_X v_\epsilon = \{y\}$ and

$$a(v_\epsilon, X, 0) = a(v, X, 0) + e\epsilon + (d - e - r)\epsilon^2.$$

By Definition 2.1, $v_\epsilon(\bar{s}_i)$ is the minimum of

$$\ell(\gamma) + \epsilon|\beta| + \epsilon^2 \sum_{l=r+1}^{d-e} \gamma_l$$

over the exponents (γ, β) with a non-zero coefficient in the expansion of $h^*\bar{s}_i$. Only finitely many $(\gamma_1, \dots, \gamma_r)$ have $\ell(\gamma)$ below a given bound, so there exists a real number $\delta > 0$ such that for any i, any such exponent (γ, β) with $\ell(\gamma) > \ell(\gamma^{(i)})$ satisfies $\ell(\gamma) \geq \ell(\gamma^{(i)}) + \delta$. Any such exponent (γ, β) with $\ell(\gamma) = \ell(\gamma^{(i)})$ satisfies $|\beta| \geq |\beta^{(i)}|$. Thus for any sufficiently small ϵ and any i, we have

$$\ell(\gamma^{(i)}) + \epsilon|\beta^{(i)}| \leq v_\epsilon(\bar{s}_i) \leq \ell(\gamma^{(i)}) + \epsilon|\beta^{(i)}| + \epsilon^2 \sum_{l=r+1}^{d-e} \gamma_l^{(i)}.$$

Therefore, as $\epsilon \to 0^+$,

$$\psi_{m,t}(v_\epsilon, \bar{s}) - \psi_{m,t}(v, \bar{s}) = \frac{\epsilon}{N_m} \exp\left(\frac{a(v, X, 0)}{t}\right) \sum_{i=1}^{N_m} \exp\left(-\frac{\ell(\gamma^{(i)})}{m}\right) \left(\frac{e}{t} - \frac{|\beta^{(i)}|}{m}\right) + O(\epsilon^2).$$

Since $y \in \operatorname{center}_X v_\epsilon$, Construction-Proposition 5.2(1) implies that

$$\psi_{m,t}(v_\epsilon, \bar{s}) \geq \mu_{m,t}(y) \geq \mu_{m,t}(x) = \psi_{m,t}(v, \bar{s}).$$

Thus the coefficient of ϵ above is non-negative, so

$$\frac{e}{t} Q_m \geq \frac{1}{m} \sum_{(\gamma,\beta)\in\Gamma_m} |\beta| \exp\left(-\frac{\ell(\gamma)}{m}\right) = T_m Q_m.$$

Since $N_m > 0$, we have $Q_m > 0$, so $T_m \leq e/t$. □

5.2. **Growth of slices.** In this subsection, we bound T_m from below by a quantity independent of V and Y, and we prove Proposition 5.1. The method is that of [OAI26, Section 5.3]. By the Brunn–Minkowski inequality, the lower bound on I_V in Proposition 3.2(1) and the cube $\{0,1\}^e \subset G_0(ke)$ of Lemma 5.3(4) imply that for any non-empty slice $G_\gamma(j)$ and any sufficiently large integer j', the difference $b_\gamma(j+j') - b_\gamma(j)$ is at least j' minus a constant independent of V and Y, so we can let $m \to +\infty$ although V and Y vary with m.

In the proof of Proposition 5.1, the exponential factor coming from Lemma 5.3(3) can be discarded for $t > d+1$. For $t < d+1$, the proof depends on the lower bound $a_* > 0$ of Lemma 4.7(2), and we have to fix a_* before t [OAI26, (5.17) and Table 1]. See (b) in the sketch of the proof in Section 1 for the role of $L^d \geq 2$.

Lemma 5.5. *Notation as in Construction-Proposition 5.2. Let c_e be a positive integer which satisfies Proposition 3.2(1), and $c' := c_e + ke + 1$. Then for any positive integers $j_1 < \cdots < j_l = m$ with $N_{j_1} > 0$ and $j_{i+1} - j_i \geq c'$ for any i, we have*

$$T_m + \frac{e}{m} \geq e \sum_{i=1}^{l-1} \frac{j_{i+1} - j_i - c'}{m} \cdot \frac{Q_{j_i}}{Q_m}.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove a sumset inequality in $\mathbb{Z}^e$.

Let A and C be non-empty finite subsets of $\mathbb{Z}^e$. For any finite subset $P \subset \mathbb{Z}^e$, the set $P + [0,1]^e$ is a finite union of unit cubes and $\operatorname{vol}(P + [0,1]^e) = \#P$. Since $[0,2]^e = \{0,1\}^e + [0,1]^e$, we have

$$(A + [0,1]^e) + (C + [0,1]^e) \subset A + C + \{0,1\}^e + [0,1]^e.$$

Thus the Brunn–Minkowski inequality [Gar02, Theorem 4.1] (cf. [OAI26, (5.13)]), applied to $A + [0,1]^e$ and $C + [0,1]^e$, implies that

$$(\#(A + C + \{0,1\}^e))^{1/e} \geq (\#A)^{1/e} + (\#C)^{1/e}.$$

Step 2. In this step we show that the slices grow with the degree, and that $N_{j_i} > 0$ for any i.

Let j and j' be positive integers with $N_j > 0$ and $j' \geq c'$. Then $j' - ke \geq c_e + 1$. Since $y \in V \cap U$ and $\dim V = e$, Proposition 3.2(1) applied to the integer $j' - ke$ implies that

$$(e! I_V(j' - ke))^{1/e} \geq j' - ke - c_e \geq 1.$$

Thus $I_V(j' - ke) > 0$, so $N_{j'-ke} > 0$, and $G_0(j' - ke)$ is non-empty with $\#G_0(j' - ke) \geq I_V(j' - ke)$ (Lemma 5.3(4)). Since $\tau_0(y) \neq 0$, we have $\tau_0^e \neq 0$ in R_{ke}, so $N_{ke} > 0$ and $\{0,1\}^e \subset G_0(ke)$ (Lemma 5.3(4)). By Lemma 5.3(1), we have $N_{j+j'} > 0$ and

$$\Gamma_j + \Gamma_{j'-ke} + \Gamma_{ke} \subset \Gamma_{j+j'}.$$

By induction on i, $N_{j_i} > 0$ for any i.

Let $\gamma \in \mathbb{Z}_{\geq 0}^{d-e}$ with $G_\gamma(j) \neq \emptyset$. Let $A := G_\gamma(j)$ and $C := G_0(j' - ke)$. Then $A + C + \{0,1\}^e \subset G_\gamma(j + j')$. By **Step 1**,

$$b_\gamma(j + j') \geq b_\gamma(j) + b_0(j' - ke) \geq b_\gamma(j) + j' - ke - c_e \geq b_\gamma(j) + j' - c'. \tag{5.1}$$

Step 3. In this step we define a weighted sum Ω_j of powers of the $b_\gamma(j)$ and show that Ω_j grows with the degree.

For a positive integer j with $N_j > 0$, we let

$$\Omega_j := \frac{1}{e!} \sum_\gamma \exp\left(-\frac{\ell(\gamma)}{m}\right) b_\gamma(j)^{e+1},$$

where the sum runs over the finitely many γ with $G_\gamma(j) \neq \emptyset$. Let j and j' be positive integers with $N_j > 0$ and $j' \geq c'$. If $G_\gamma(j) \neq \emptyset$, then $j' - c' \geq 0$, (5.1), and the convexity of λ^{e+1} on $\mathbb{R}_{\geq 0}$ imply that

$$b_\gamma(j + j')^{e+1} \geq b_\gamma(j)^{e+1} + (e+1)(j' - c')b_\gamma(j)^e = b_\gamma(j)^{e+1} + (e+1)!(j' - c')\#G_\gamma(j).$$

We have

$$Q_j = \sum_\gamma \#G_\gamma(j) \exp\left(-\frac{\ell(\gamma)}{m}\right),$$

and $\exp(-\ell(\gamma)/m) b_\gamma(j + j')^{e+1} \geq 0$ for any γ with $G_\gamma(j) = \emptyset$. Thus

$$\Omega_{j+j'} \geq \Omega_j + (e+1)(j' - c')Q_j. \tag{5.2}$$

Step 4. In this step we bound the sum of the $|\beta|$ from below by Ω_m.

Let $A \subset \mathbb{Z}_{\geq 0}^e$ be a non-empty finite subset and $b := (e!\#A)^{1/e}$. Let $w_1, \ldots, w_e$ be the coordinates of $\mathbb{R}^e$, $f := \sum_i w_i$,

$$S := A + [0,1]^e \subset \mathbb{R}_{\geq 0}^e, \quad \Delta := \left\{ w \in \mathbb{R}_{\geq 0}^e \middle| \sum_i w_i \leq b \right\}.$$

Then $\operatorname{vol} S = \#A = b^e/e! = \operatorname{vol} \Delta$. Since $f \leq b$ on Δ and $f > b$ on $\mathbb{R}_{\geq 0}^e \backslash \Delta$, we have (cf. [LL01, Theorem 1.14])

$$\int_S f - \int_\Delta f = \int_{S \backslash \Delta} f - \int_{\Delta \backslash S} f \geq b \operatorname{vol}(S \backslash \Delta) - b \operatorname{vol}(\Delta \backslash S) = 0.$$

A direct calculation shows that

$$\int_\Delta f = \frac{eb^{e+1}}{(e+1)e!}, \quad \int_S f = \sum_{\beta \in A} \left(|\beta| + \frac{e}{2}\right).$$

Thus

$$\sum_{\beta \in A} (|\beta| + e) \geq \sum_{\beta \in A} \left(|\beta| + \frac{e}{2}\right) \geq \frac{e}{e+1} \cdot \frac{b^{e+1}}{e!}.$$

Let $A := G_\gamma(m)$ for any γ with $G_\gamma(m) \neq \emptyset$. Multiplying by $\exp(-\ell(\gamma)/m)$ and summing over γ, we obtain

$$\sum_{(\gamma,\beta) \in \Gamma_m} \exp\left(-\frac{\ell(\gamma)}{m}\right) (|\beta| + e) \geq \frac{e}{e+1} \Omega_m. \tag{5.3}$$

Step 5. In this step we conclude the proof.

For $1 \leq i \leq l - 1$, (5.2) with $j = j_i$ and $j' = j_{i+1} - j_i$ implies that

$$\Omega_{j_{i+1}} \geq \Omega_{j_i} + (e+1)(j_{i+1} - j_i - c')Q_{j_i}.$$

Since $j_l = m$ and $\Omega_{j_1} \geq 0$, summing over i we obtain

$$\Omega_m \geq (e+1)\sum_{i=1}^{l-1}(j_{i+1} - j_i - c')Q_{j_i}.$$

By the definitions of Q_m and T_m, the left-hand side of (5.3) is equal to $mQ_m(T_m + e/m)$. Thus (5.3) implies that

$$mQ_m\left(T_m + \frac{e}{m}\right) \geq e\sum_{i=1}^{l-1}(j_{i+1} - j_i - c')Q_{j_i}.$$

Since $Q_m > 0$, dividing by mQ_m concludes the proof. □

Proof of Proposition 5.1. The self-intersection number L^d is a positive integer ([Laz04, Theorem 2.2.16]). We fix positive real numbers $t_- \leq t_+$ as follows. For (1), let t be a real number with $t > d+1$, and let $t_- = t_+ = t$. Then $t_-(L^d)^{1/d} \geq t > d+1$. For (2), let $t_+ := d+1$, and since $L^d \geq 2$, we fix a positive real number $t_- < d+1$ with $t_-(L^d)^{1/d} > d+1$.

Lemma 4.7 applied to x, t_-, and t_+ implies that there exist real numbers $\delta \in (0,1)$, $\kappa > 0$, $a_* > 0$, $a^* > 0$, and a positive integer m_0 for which (1)–(3) of Lemma 4.7 hold. Let

$$c_* := \int_0^1 \lambda^d \exp\left((1-\lambda)\frac{a_*}{t_+}\right) d\lambda.$$

Since $a_* > 0$, we have $c_* > 1/(d+1)$. For (1), $c_* > 1/(d+1) > 1/t$. For (2), c_* is now fixed and $c_* > 1/(d+1)$, so we may fix a real number $t \in [t_-, d+1)$ with $c_* > 1/t$. Thus in (2), a_* is fixed before t.

Let $\mu_* := \liminf_{j\to+\infty} \mu_{j,t}(x)$. By Lemma 4.7(1), $\mu_* \geq \kappa > 0$. Let Λ be an infinite set of integers $m \geq m_0$ such that $\mu_{m,t}(x) \to \mu_*$ as $m \to +\infty$ in Λ.

Suppose that there exists an infinite subset $\Lambda' \subset \Lambda$ such that for any $m \in \Lambda'$, $\psi_{m,t}$ has a minimizer (v,s) at x with $\operatorname{center}_X v$ not a point. Let k be a positive integer which satisfies Proposition 3.2(2). For any $m \in \Lambda'$, Construction-Proposition 5.2 implies that $e := \dim \operatorname{center}_X v$ satisfies $1 \leq e \leq d-1$. Possibly replacing Λ' with an infinite subset, we may assume that e does not depend on m. We fix a positive integer c_e which satisfies Proposition 3.2(1), and we let $c' := c_e + ke + 1$. For any $m \in \Lambda'$, we take the data (1)–(9) of Construction-Proposition 5.2. The variety Y and the data (1)–(9) vary with m, while c' does not.

Fix an integer $l \geq 2$, and let $j_i := \lfloor mi/l \rfloor$ for $1 \leq i \leq l$. Then $j_l = m$. For any sufficiently large $m \in \Lambda'$, we have $j_{i+1} - j_i \geq \lfloor m/l \rfloor - 1 \geq c'$ for any i and $N_{j_1} > 0$. Lemma 5.5 and Lemma 5.3(3) imply that

$$T_m + \frac{e}{m} \geq e\sum_{i=1}^{l-1} \frac{j_{i+1} - j_i - c'}{m} \cdot \frac{N_{j_i}}{N_m} \cdot \frac{\mu_{j_i,t}(x)}{\mu_{m,t}(x)} \exp\left(\left(1 - \frac{j_i}{m}\right)\frac{a(v,X,0)}{t}\right).$$

Since $\psi_{m,t}(v,s) = \mu_{m,t}(x) \leq 1-\delta$ (Lemma 4.7(1)), we have $a(v,X,0) \geq a_*$ (Lemma 4.7(2)). Moreover, $t \leq t_+$. As $m \to +\infty$ in Λ', we have $(j_{i+1} - j_i - c')/m \to 1/l$, $j_i/m \to i/l$, $N_{j_i}/N_m \to (i/l)^d$ ([Laz04, Corollary 1.4.41] and $L^d > 0$), and $\liminf \mu_{j_i,t}(x)/\mu_{m,t}(x) \geq 1$ by the definition of μ_*. Thus

$$\liminf_{m\to+\infty, m\in\Lambda'} T_m \geq e\sum_{i=1}^{l-1} \frac{1}{l}\left(\frac{i}{l}\right)^d \exp\left(\left(1 - \frac{i}{l}\right)\frac{a_*}{t_+}\right).$$

Letting $l \to +\infty$, the right-hand side being a Riemann sum of the integral defining c_*, we obtain

$$\liminf_{m\to+\infty, m\in\Lambda'} T_m \geq ec_* > \frac{e}{t},$$

which contradicts Lemma 5.4.

Thus for all but finitely many $m \in \Lambda$, the center of any minimizer of $\psi_{m,t}$ at x is a point. For any such m, $\psi_{m,t}$ has a minimizer at x (Lemma 4.7(3)), and any minimizer (v,s) of $\psi_{m,t}$ at x

satisfies $x \in \operatorname{center}_X v$, so $\operatorname{center}_X v = \{x\}$. Therefore (m,t) is good at x for infinitely many positive integers m. □

6. Separation of points

The goal of this section is to prove Proposition 6.1, from which Theorems 1.5 and 1.6 will both follow.

Proposition 6.1. *Notation and conditions as in Set-up 3.1. Let q be a rational number, M a Cartier divisor on X, and $F_0 \geq 0$ a $\mathbb{Q}$-divisor on X with $M - qL \sim_{\mathbb{Q}} F_0$. Assume that $q > d+1$, or that $q \geq d+1$ and $L^d \geq 2$. Then $|K_X + M|$ defines a birational map onto the image.*

Section 5 only considers one closed point x. For a very general closed point x of U, by Proposition 5.1, there exists a pair (m,t) good at x, and by Lemma 4.4(3), x belongs to U_m°. In Proposition 6.5, for any other closed point y of U, we use such a pair to construct a section of $K_X + M$ which is non-zero at x and zero at y. For y, no further positivity of L is used (see (d) in the sketch of the proof in Section 1).

Set-up 6.2. Notation and conditions as in Set-up 3.1. Let m, t, x, and y be as follows.

(1) m is a positive integer such that $|mL|$ has no base point in U and separates any two distinct closed points of U.
(2) t is a positive real number and $x \in U_m^\circ$ is a closed point such that (m,t) is good at x.
(3) $y \in U \backslash \{x\}$ is a closed point.

We do not assume that (m,t) is good at y, and we do not assume a minimizer at y. The point y may belong to the fiber through x of the rational map defined by $|K_X + M|$, where M is the divisor in Proposition 6.1, and this is the case which we have to exclude.

6.1. A strict inequality.

In this subsection, we prove the strict inequality of Construction-Proposition 6.3(2).

The numbers b_i in Construction-Proposition 6.3 come from [OAI26, Section 6.1]. For Construction-Proposition 6.3(2), see (d) in the sketch of the proof in Section 1 (cf. [Nug26, Lemma 3.1]).

Construction-Proposition 6.3. Notation and conditions as in Set-up 6.2. Then there exist an ordered basis s of R_m, a log resolution $h : Y \to X$ of $(X, \sum_i D_i)$ with $D_i := \operatorname{div}(s_i)$, a monomial valuation v on $(Y, \sum_j E_j)$ with $\sum_j E_j$ the reduced divisor on Y with support $\operatorname{Exc}(h) \cup \operatorname{Supp} h^* \sum_i D_i$, and a prime divisor E_y over X with $y \in \operatorname{center}_X E_y$, such that (v,s) is a minimizer of $\psi_{m,t}$ at x and the real numbers

$$b_i := \frac{t}{m} \exp\left(-\frac{v(s_i)}{m}\right) \left(\sum_{i'} \exp\left(-\frac{v(s_{i'})}{m}\right)\right)^{-1}$$

satisfy the following.

(1) $\sum_i b_i v(s_i) = a(v, X, 0)$.
(2) $\sum_i b_i \operatorname{mult}_{E_y} D_i > a(E_y, X, 0)$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix a minimizer at x whose basis is adapted to the valuation.

Since L is big, we have $U \neq \emptyset$, so Set-up 6.2(1) implies that $N_m > 0$. Since (m,t) is good at x, there exists a minimizer $(v, \bar{s})$ of $\psi_{m,t}$ at x. By Lemma 4.2(2) with $w = v$, there exists an ordered basis s^x of R_m adapted to v. By Lemma 4.2(1),

$$\psi_{m,t}(v, s^x) \leq \psi_{m,t}(v, \bar{s}) = \mu_{m,t}(x).$$

Thus (v, s^x) is a minimizer of $\psi_{m,t}$ at x, and $\operatorname{center}_X v = \{x\}$ because (m,t) is good at x. Possibly renumbering s^x, we may assume that the numbers $p_i^\downarrow := v(s_i^x)/a(v, X, 0)$ satisfy $p_1^\downarrow \geq \cdots \geq p_{N_m}^\downarrow$.

We have $p_1^\downarrow > 0$. Otherwise $v(s_i^x) = 0$ for any i, and for any real number $c \in (0,1)$, the monomial valuation cv satisfies

$$\psi_{m,t}(cv, s^x) = \exp\left(\frac{ca(v,X,0)}{t}\right) < \exp\left(\frac{a(v,X,0)}{t}\right) = \psi_{m,t}(v, s^x),$$

which contradicts the minimality of (v, s^x).

Step 2. In this step we realize the numbers $p_i^\downarrow$ at y by a monomial valuation.

Let α be any decreasing list with $\alpha_i < p_i^\downarrow$ for any i with $p_i^\downarrow > 0$ and $\alpha_i = 0$ for any i with $p_i^\downarrow = 0$. By Lemma 2.2(1) applied to v and $\mathcal{S} = \{s_1^x, \dots, s_{N_m}^x\}$, there exists a divisorial valuation v_0 over X with $\operatorname{center}_X v_0 = \{x\}$ and $v_0(s_i^x) \geq \alpha_i a(v_0, X, 0)$ for any i. Thus x realizes α in degree m. Since $x \in U_m^\circ$, by Lemma 4.4(3), y realizes α in degree m.

Thus the assumption of Lemma 4.5(2) holds with $z = y$ and $p = p^\downarrow$. By Lemma 4.5, there exist an ordered basis s^y of R_m and a monomial valuation w over X such that $y \in \operatorname{center}_X w$, $a(w, X, 0) = 1$, and $w(s_i^y) \geq p_i^\downarrow$ for any i.

Step 3. In this step we construct s and the numbers b_i, and we prove the equality (1) for v.

By Lemma 4.2(2), there exists an ordered basis s of R_m adapted to both v and w (cf. [Nug26, Lemma 2.1]). Since s and s^x are both adapted to v, we may use Lemma 4.2(1) twice: there exist permutations ρ and ρ' of $\{1, \dots, N_m\}$ such that

$$v(s_{\rho(i)}) \geq v(s_i^x) \quad \text{and} \quad v(s^x_{\rho'(i)}) \geq v(s_i)$$

for any i. Summing over i, we obtain $\sum_i v(s_i) = \sum_i v(s_i^x)$, so $v(s_{\rho(i)}) = v(s_i^x)$ for any i. Thus the numbers

$$p_i := \frac{v(s_i)}{a(v,X,0)}$$

form a permutation $p = (p_i)$ of $p^\downarrow$, and (v, s) is a minimizer of $\psi_{m,t}$ at x.

Let $p_i' := w(s_i)$ and $p' = (p_i')$, and let $(p')^\downarrow$ be the decreasing rearrangement of p'. Since s is adapted to w, by Lemma 4.2(1) applied to w, s, and s^y, there exists a permutation ρ'' of $\{1, \dots, N_m\}$ such that

$$w(s_{\rho''(i)}) \geq w(s_i^y) \geq p_i^\downarrow$$

for any i. Thus for any i, at least i of the numbers p_j' are $\geq p_i^\downarrow$, that is,

$$(p')_i^\downarrow \geq p_i^\downarrow \quad \text{for any } i. \tag{6.1}$$

Let b_i be the numbers defined in the statement. Then $b_i > 0$ and $m\sum_i b_i = t$. Since $\operatorname{center}_X(cv) = \operatorname{center}_X v$ for any real number $c > 0$ (Definition 2.1), the function

$$c \to \psi_{m,t}(cv, s) = \frac{1}{N_m}\sum_i \exp\left(c\left(\frac{a(v,X,0)}{t} - \frac{v(s_i)}{m}\right)\right)$$

has a minimum at $c = 1$, so the derivative at $c = 1$ is zero. Dividing by $\exp(a(v,X,0)/t)/N_m$, we obtain

$$\sum_i \left(\frac{a(v,X,0)}{t} - \frac{v(s_i)}{m}\right)\exp\left(-\frac{v(s_i)}{m}\right) = 0.$$

Multiplying by t/m and dividing by $\sum_{i'} \exp(-v(s_{i'})/m)$, we obtain

$$\frac{a(v,X,0)}{t}\sum_i b_i = \frac{1}{m}\sum_i b_i v(s_i).$$

Since $m\sum_i b_i = t$, we have $\sum_i b_i v(s_i) = a(v,X,0)$, that is, $\sum_i b_i p_i = 1$. Moreover, since $v(s_i) = a(v,X,0)p_i$, for any i and j, we have $b_i < b_j$ if $p_i > p_j$, and $b_i = b_j$ if $p_i = p_j$.

Step 4. In this step we prove that $\sum_i b_i p_i' > 1$.

Let π be a permutation of $\{1,\dots,N_m\}$ with $p_{\pi(i)} = p_i^{\downarrow}$ for any i, and let $b_i^{\uparrow} := b_{\pi(i)}$. By **Step 3**, $b_1^{\uparrow} \le \cdots \le b_{N_m}^{\uparrow}$, and

$$\sum_i b_i p_i' \ge \sum_i b_i^{\uparrow} (p')_i^{\downarrow} \ge \sum_i b_i^{\uparrow} p_i^{\downarrow} = \sum_i b_i p_i = 1. \tag{6.2}$$

The first inequality is [HLP52, Theorem 368], and the second inequality follows from (6.1) and $b_i^{\uparrow} > 0$.

Suppose that $\sum_i b_i p_i' = 1$. Then both inequalities in (6.2) are equalities. We first prove that $p' = p$.

Since $b_i^{\uparrow} > 0$ for any i, equality in the second inequality and (6.1) imply that $(p')^{\downarrow} = p^{\downarrow}$, so p' is a permutation of p. Since the first inequality of (6.2) holds for any permutation of p' in place of p', equality in the first inequality implies that $\sum_i b_i p_i'$ is the smallest among the values of $\sum_i b_i p_i''$ for all permutations p'' of p'. If $p_i > p_j$ and $p_i' < p_j'$, then $b_i < b_j$, and exchanging p_i' and p_j' decreases the sum by

$$(b_j - b_i)(p_j' - p_i') > 0,$$

which is not possible. Thus $p_i > p_j$ implies that $p_i' \ge p_j'$ for any i and j.

Let λ be the maximum of the p_i, and let I be the set of indices i with $p_i = \lambda$. For any $i \in I$ and $j \notin I$, we have $p_i' \ge p_j'$. Since p' is a permutation of p, exactly $\#I$ of the numbers p_i' are equal to λ, and λ is the maximum of the p_i'. Thus $p_i' = \lambda$ for any $i \in I$. Removing the indices in I and inducting on the number of distinct values of p, we obtain $p' = p$.

We now compare the kernels of the evaluation maps of R_m at x and y. By the definition of v, we have $v(\sigma+\sigma') \ge \min\{v(\sigma), v(\sigma')\}$ for any non-zero $\sigma, \sigma', \sigma+\sigma' \in R_m$, and a non-zero $\sigma \in R_m$ vanishes on $\operatorname{center}_X v$ if and only if $v(\sigma) > 0$. Both properties hold with w in place of v. Let R' be the subspace of R_m generated by the s_i with $p_i > 0$.

Let $\sigma \in R_m$ be a non-zero section. If $v(\sigma) > 0$, then since s is adapted to v, σ is a linear combination of the s_i with $p_i > 0$, so $\sigma \in R'$. Conversely, if $\sigma \in R'$, then $v(\sigma) \ge \min_{p_i>0} v(s_i) > 0$. Thus $v(\sigma) > 0$ if and only if $\sigma \in R'$. Since s is adapted to w and $w(s_i) = p_i' = p_i$ for any i, we also have $w(\sigma) > 0$ if and only if $\sigma \in R'$.

Let e_x and e_y be the evaluation maps of R_m at x and y. Since $\operatorname{center}_X v = \{x\}$, we have $R' = \operatorname{Ker} e_x$. Since $y \in \operatorname{center}_X w$, we have $R' \subset \operatorname{Ker} e_y$. Since $|mL|$ has no base point in U, both $\operatorname{Ker} e_x$ and $\operatorname{Ker} e_y$ have codimension one in R_m. Thus $\operatorname{Ker} e_x = \operatorname{Ker} e_y$, which contradicts the assumption that $|mL|$ separates x and y.

Thus, by (6.2),

$$\sum_i b_i w(s_i) = \sum_i b_i p_i' > 1 = a(w, X, 0). \tag{6.3}$$

Step 5. In this step we construct E_y and prove the inequality (2).

Let $D_i := \operatorname{div}(s_i)$. By Lemma 2.2(1) applied to w and $\mathcal{S} = \{s_1, \dots, s_{N_m}\}$, and by (6.3), there exists a divisorial valuation w_0 over X with $\operatorname{center}_X w_0 = \operatorname{center}_X w$ and

$$\sum_i b_i w_0(s_i) > a(w_0, X, 0).$$

By Definition 2.1, there exist a prime divisor E_y over X and a real number $c > 0$ with $w_0(D) = c \operatorname{mult}_{E_y} D$ for any Cartier divisor $D \ge 0$ on X. Then $y \in \operatorname{center}_X E_y$. Since $w_0(s_i) = c \operatorname{mult}_{E_y} D_i$ and $a(w_0, X, 0) = c a(E_y, X, 0)$, dividing by c, we obtain

$$\sum_i b_i \operatorname{mult}_{E_y} D_i > a(E_y, X, 0).$$

Step 6. In this step we replace v with a monomial valuation on a log resolution of $(X, \sum_i D_i)$.

Let $h : Y \to X$ be a log resolution of $(X, \sum_i D_i)$, and let $\sum_j E_j$ be the reduced divisor on Y with support $\operatorname{Exc}(h) \cup \operatorname{Supp} h^* \sum_i D_i$. Recall that $p_i = v(s_i)/a(v, X, 0)$, that $p_i \ge 0$ for any i, and

that $p_i > 0$ for some i, by **Steps 1 and 3**. By Lemma 2.2(1) applied to v and $\mathcal{S} = \{s_1, \dots, s_{N_m}\}$, the assumption of Lemma 4.5(1) holds with $z = x$ and these p_i. By Lemma 4.5(1), there exists a monomial valuation w' on $(Y, \sum_j E_j)$ with $x \in \operatorname{center}_X w'$, $a(w', X, 0) = 1$, and $w'(s_i) \geq p_i$ for any i.

Let $v' := a(v, X, 0)w'$. Then v' is a monomial valuation on $(Y, \sum_j E_j)$ with $x \in \operatorname{center}_X v'$, $a(v', X, 0) = a(v, X, 0)$, and $v'(s_i) \geq v(s_i)$ for any i. Thus

$$\psi_{m,t}(v', s) \leq \psi_{m,t}(v, s) = \mu_{m,t}(x).$$

By the definition of $\mu_{m,t}(x)$, equality holds. Since each summand of $\psi_{m,t}(v', s)$ is strictly decreasing in $v'(s_i)$, we have $v'(s_i) = v(s_i)$ for any i. Thus (v', s) is a minimizer of $\psi_{m,t}$ at x, the numbers b_i do not change, and (1) and (2) hold with v' in place of v. Possibly replacing v with v', the pair (v, s), the morphism h, and the prime divisor E_y satisfy all the conclusions. □

6.2. **The boundary.** In this subsection, we construct the boundary of Lemma 6.4. The constructions in Lemma 6.4 and Proposition 6.5 are those of [OAI26, Sections 6.1–6.3]. For ample L, they also appear in [Nug26, Propositions 6.2 and 6.3], which go back to [OAI26] (see [Nug26, Appendix A, item (5)]). The mechanism of the lifting is [Fuj93, Section 1, (1.6)].

We note the following differences. First, L is ample in [OAI26] and [Nug26], while here L is only nef and big, so the global generation of the ideal uses a divisor of Lemma 2.4(1) which does not vanish at x. The choice in [OAI26, Section 6.2] of a twist of the ideal by a multiple of L uses Serre's theorem, which requires L ample: on the blow-up of $\mathbb{P}^2$ at a point, the ideal sheaf of a point on the exceptional curve is not globally generated after a twist by any multiple of the pullback of a line, since the global sections of such a twist are pulled back from $\mathbb{P}^2$ and hence vanish along the exceptional curve once they vanish at the point. Second, [OAI26, Section 6] considers one point x and has no counterpart of the strict inequality in Lemma 6.4(4). In [Nug26, Proposition 6.3], that inequality is a hypothesis, which follows from [Nug26, Lemma 3.1] and the continuity argument [Nug26, (7.1)]. Here it follows from Construction-Proposition 6.3(2). Third, the $\mathbb{Q}$-divisor F_0 of Lemma 6.4 has no counterpart in [OAI26, Sections 6.1–6.3] or [Nug26, Propositions 6.2 and 6.3]. Here the SNC divisor of Lemma 6.4 contains the support of the pullback of F_0 from the start, and Proposition 6.5 assumes only that the support of F_0 does not contain x.

The numbers $v(s_i)/a(v, X, 0)$ of Construction-Proposition 6.3 are real, while the numbers $\operatorname{mult}_F D_i/a(F, X, 0)$ for a prime divisor F over X are rational, and Lemma 6.4(2) requires prime divisors with coefficient equal to the log discrepancy, so we use [OAI26, Lemma 6.1] to replace the numbers $v(s_i)/a(v, X, 0)$ and b_i with rational numbers.

Lemma 6.4 (cf. [OAI26, Lemma 6.2] and [Nug26, Proposition 6.2])**.** *Notation and conditions as in Set-up 6.2. Let $q > t$ be a rational number and $F_0 \geq 0$ a $\mathbb{Q}$-divisor on X with $x \notin \operatorname{Supp} F_0$. Then there exist a birational morphism $g : Y' \to X$ from a smooth projective variety Y', a $\mathbb{Q}$-divisor $B \geq 0$ on Y', a non-zero reduced divisor S on Y', and a prime divisor E_y on Y' such that $\operatorname{Exc}(g) \cup \operatorname{Supp}(B + g^*F_0 + S + E_y)$ is the support of a reduced SNC divisor Σ, and the following hold.*

(1) $qg^*L - B$ is nef and big.
(2) For any component F of S, we have $g(F) = \{x\}$ and $\operatorname{mult}_F B = a(F, X, 0)$.
(3) For any component F of Σ which is not a component of S and with $F \cap S \neq \emptyset$, we have $\operatorname{mult}_F B < a(F, X, 0)$.
(4) $y \in g(E_y)$ and $\operatorname{mult}_{E_y} B > a(E_y, X, 0)$.

Proof. The proof is long, so we divide it into several steps.

Let s, D_i, $h : Y \to X$, $\sum_j E_j$, v, E_y, and b_i be as in Construction-Proposition 6.3. We let $b := (b_i)_i$, and for $e, e' \in \mathbb{R}^{N_m}$, we let $e \cdot e' := \sum_i e_i e'_i$. We let

$$a_j := a(E_j, X, 0), \qquad \mathbf{z}_j := \left(\frac{\operatorname{mult}_{E_j} D_i}{a_j}\right)_i, \qquad p := \left(\frac{v(s_i)}{a(v, X, 0)}\right)_i.$$

Step 1. In this step we construct rational vectors p^0 and b^0 and a prime divisor F_* over X with $\operatorname{center}_X F_* = \{x\}$.

For a stratum W of $\sum_j E_j$ with $x \in h(W)$, we let

$$C_W := \operatorname{Conv}\{\mathbf{z}_j \mid W \subset E_j\}.$$

The C_W form a finite set of rational polytopes in $\mathbb{R}^{N_m}_{\geq 0}$.

Let W be a stratum of $\sum_j E_j$ with $x \in h(W)$ and $\lambda_j \geq 0$ real numbers with $\sum_j \lambda_j = 1$ and $\lambda_j = 0$ unless $W \subset E_j$. Let W' be the irreducible component of $\bigcap_{\lambda_j > 0} E_j$ with $W \subset W'$, and v_λ the monomial valuation on $(Y, \sum_j E_j)$ with weights $a(v, X, 0)\lambda_j / a_j$ along the E_j with $\lambda_j > 0$, at W'. Since $h^* D_i = \sum_j (\operatorname{mult}_{E_j} D_i) E_j$, we have

$$a(v_\lambda, X, 0) = a(v, X, 0), \qquad \left(\frac{v_\lambda(s_i)}{a(v, X, 0)}\right)_i = \sum_j \lambda_j \mathbf{z}_j, \qquad \operatorname{center}_X v_\lambda = h(W'). \tag{6.4}$$

Since (m, t) is good at x and (v, s) is a minimizer of $\psi_{m,t}$ at x, we have $\operatorname{center}_X v = \{x\}$. Thus $x \in h(W_v)$, where W_v is the stratum at which v is defined. Let $\lambda_j := 0$ if $W_v \not\subset E_j$, and otherwise let $a(v, X, 0)\lambda_j / a_j$ be the weight of v along E_j. Then $\sum_j \lambda_j = 1$ and v is the monomial valuation v_λ with $W = W_v$, so (6.4) implies that $p = \sum_j \lambda_j \mathbf{z}_j$. Thus $p \in C_{W_v}$. If $p \in C_W$ with $\dim h(W) > 0$, then we write $p = \sum_j \lambda_j \mathbf{z}_j$ with $\lambda_j \geq 0$, $\sum_j \lambda_j = 1$, and $\lambda_j = 0$ unless $W \subset E_j$, and (6.4) implies that (v_λ, s) is a minimizer of $\psi_{m,t}$ at x with $\dim \operatorname{center}_X v_\lambda > 0$. This contradicts the assumption that (m, t) is good at x. Thus $p \notin C_W$ for any W with $\dim h(W) > 0$.

Assume that $p \in C_W$ and let $p' \in C_W$. For $\tau \in [0, 1]$, let $p_\tau := (1 - \tau)p + \tau p' \in C_W$, and write $p_\tau = \sum_j \lambda_j \mathbf{z}_j$ with $\lambda_j \geq 0$, $\sum_j \lambda_j = 1$, and $\lambda_j = 0$ unless $W \subset E_j$. By (6.4),

$$\frac{1}{N_m} \sum_i \exp\left(\frac{a(v, X, 0)}{t} - \frac{a(v, X, 0)(p_\tau)_i}{m}\right) = \psi_{m,t}(v_\lambda, s) \geq \mu_{m,t}(x) = \psi_{m,t}(v, s).$$

The left-hand side is equal to $\psi_{m,t}(v, s)$ for $\tau = 0$, so the derivative of the left-hand side at $\tau = 0$ is non-negative. Since $v(s_i) = a(v, X, 0)p_i$, we obtain

$$\sum_i \exp\left(-\frac{v(s_i)}{m}\right)(p'_i - p_i) \leq 0.$$

By the formula for b_i and Construction-Proposition 6.3(1), we obtain

$$b \cdot p' \leq b \cdot p = 1.$$

By the formula for b_i, we have $b_i > 0$ for any i and $m \sum_i b_i = t$. Since $q > t$, and by Construction-Proposition 6.3(2), b belongs to the open subset

$$\mathcal{V} := \left\{ b' \in \mathbb{R}^{N_m}_{>0} \;\middle|\; m \sum_i b'_i < q,\ \sum_i b'_i \operatorname{mult}_{E_y} D_i > a(E_y, X, 0) \right\}$$

of $\mathbb{R}^{N_m}$. By [OAI26, Lemma 6.1], applied to the set of the C_W, the point p, the vector b, and $\mathcal{V}$, there exist $p^0 \in \mathbb{Q}^{N_m}$ and $b^0 = (b^0_i)_i \in \mathcal{V} \cap \mathbb{Q}^{N_m}_{>0}$ such that

(i) for any W, $p^0 \in C_W$ if and only if $p \in C_W$, and
(ii) $b^0 \cdot p^0 = 1$, and $b^0 \cdot e \leq 1$ for any W with $p^0 \in C_W$ and any $e \in C_W$.

By (i), $p^0 \in C_{W_v} \subset \mathbb{R}^{N_m}_{\geq 0}$, and $p^0 \notin C_W$ for any W with $\dim h(W) > 0$. By (ii), $p^0 \neq 0$.

Let W be a stratum with $x \in h(W)$ and $p^0 \in C_W$, and write $p^0 = \sum_j \lambda_j \mathbf{z}_j$ with rational $\lambda_j \geq 0$, $\sum_j \lambda_j = 1$, and $\lambda_j = 0$ unless $W \subset E_j$ (possible since p^0 and the $\mathbf{z}_j$ are rational: a rational point of a rational polytope is a rational convex combination of the vertices of the polytope, see [OAI26, Section 6.1]). Let W' be the irreducible component of $\bigcap_{\lambda_j > 0} E_j$ with $W \subset W'$. Then $p^0 \in C_{W'}$ and $x \in h(W')$, so $h(W') = \{x\}$. Let c be a positive rational number

such that the $c\lambda_j/a_j$ with $\lambda_j > 0$ are coprime positive integers. Then there exists a prime divisor F_* over X with $\operatorname{mult}_{F_*} E_j = c\lambda_j/a_j$ for any j, and

$$a(F_*, X, 0) = c, \qquad \left(\frac{\operatorname{mult}_{F_*} D_i}{a(F_*, X, 0)}\right)_i = p^0, \qquad \operatorname{center}_X F_* = \{x\},$$

see [OAI26, Section 2.1, formula (2.2) and the second paragraph after it] and [KM98, Lemma 2.45].

Step 2. In this step we construct $g : Y' \to X$ and an ideal $\mathfrak{c}$ on X with $\mathfrak{c}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-G)$.

Let $\mathcal{I} := \{i \mid p_i^0 > 0\}$, let n be a positive integer such that n/p_i^0 is an integer for any $i \in \mathcal{I}$, and let

$$\mathfrak{c} := \sum_{i\in\mathcal{I}} \mathcal{O}_X\left(-\frac{n}{p_i^0} D_i\right).$$

For any prime divisor F over X, we have

$$\frac{\operatorname{mult}_F \mathfrak{c}}{n} = \min_{i\in\mathcal{I}} \frac{\operatorname{mult}_F D_i}{p_i^0}. \tag{6.5}$$

Let A be a very ample Cartier divisor on X such that $\mathfrak{c} \otimes \mathcal{O}_X(A)$ is globally generated. Since $x \in U$, by Lemma 2.4(1), there exist a positive integer u and $D' \in |uL - A|$ with $x \notin \operatorname{Supp} D'$.

By [KM98, Theorem 0.2], applied, after passing by [KM98, Lemma 2.45] to a projective model of Y on which F_* and E_y are divisors, to the product of $\mathfrak{c}$ and the ideals of F_*, E_y, the exceptional locus over X, and the support of the pullback of $\sum_i D_i + D' + F_0$, there exist a smooth projective variety Y', a birational morphism $f : Y' \to Y$, and a Cartier divisor $G \geq 0$ on Y' such that, with $g := h \circ f$, the following hold: F_* and E_y are divisors on Y', $\mathfrak{c}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-G)$, and the reduced divisor Σ on Y' with support $\operatorname{Exc}(g) \cup \operatorname{Supp} g^*(\sum_i D_i + D' + F_0)$ is SNC. Then $g(F_*) = \{x\}$. Since $\mathfrak{c} \otimes \mathcal{O}_X(A)$ is globally generated, $g^*A - G$ is globally generated. Since $D' \sim uL - A$, we have $ug^*L - G - g^*D' \sim g^*A - G$. Thus $ug^*L - G - g^*D'$ is nef.

Step 3. In this step we construct B and prove (1) and (4).

Since $b^0 \in \mathcal{V}$, there exists a rational number $\eta \in (0, 1)$ with

$$q - (1-\eta)m\sum_i b_i^0 - \frac{\eta u}{n} > 0 \qquad \text{and} \qquad (1-\eta)\sum_i b_i^0 \operatorname{mult}_{E_y} D_i > a(E_y, X, 0).$$

Let

$$B := (1-\eta)\sum_i b_i^0 g^* D_i + \frac{\eta}{n}(G + g^*D').$$

Then $B \geq 0$. Since $D_i \sim mL$, we have

$$qg^*L - B \equiv \left(q - (1-\eta)m\sum_i b_i^0 - \frac{\eta u}{n}\right) g^*L + \frac{\eta}{n}(ug^*L - G - g^*D').$$

Since g^*L is nef and big and $ug^*L - G - g^*D'$ is nef, $qg^*L - B$ is nef and big. This proves (1).

We have $y \in \operatorname{center}_X E_y = g(E_y)$. Since $G + g^*D' \geq 0$,

$$\operatorname{mult}_{E_y} B \geq (1-\eta)\sum_i b_i^0 \operatorname{mult}_{E_y} D_i > a(E_y, X, 0).$$

This proves (4).

Since $b_i^0 > 0$ for any i and $0 < \eta < 1$, we have $\operatorname{Supp} g^* \sum_i D_i \subset \operatorname{Supp} B$. Since $\mathfrak{c} \supset \mathcal{O}_X(-(n/p_i^0)D_i)$ for $i \in \mathcal{I} \neq \emptyset$, we have $\operatorname{Supp} G \subset \operatorname{Supp} g^* \sum_i D_i$. Thus

$$\operatorname{Supp} B = \operatorname{Supp} g^*\Big(\sum_i D_i + D'\Big).$$

By Construction-Proposition 6.3(2) and $a(E_y, X, 0) > 0$, we have $E_y \subset \operatorname{Supp} g^* \sum_i D_i$. Thus the support of Σ is $\operatorname{Exc}(g) \cup \operatorname{Supp}(B + g^*F_0 + E_y)$.

Step 4. In this step we construct S and prove (2) and (3).

Let F be a component of Σ, and let

$$a_F := a(F, X, 0), \qquad r_F := \sum_j (\operatorname{mult}_F f^* E_j) a_j.$$

If $\operatorname{mult}_F f^* E_j = 0$ for all j, then $r_F = 0 < a_F$. Otherwise, let w be the divisorial valuation over X with prime divisor F and constant 1 in Definition 2.1, so $w(E_j) = \operatorname{mult}_F f^* E_j$ for any j, and let w' be the monomial valuation of Lemma 2.2(3) for w, $\sum_i D_i$, and h. Then w' is defined at a stratum W_F of $\sum_j E_j$, and

$$\begin{gathered} a(w', X, 0) = r_F \le a_F, \qquad w'(D_i) = w(D_i) = \operatorname{mult}_F D_i, \\ g(F) \subset \operatorname{center}_X w' = h(W_F). \end{gathered} \tag{6.6}$$

In particular, $r_F \le a_F$ in any case. If $r_F > 0$, then since $h^* D_i = \sum_j (\operatorname{mult}_{E_j} D_i) E_j$, we have

$$\left(\frac{\operatorname{mult}_F D_i}{r_F} \right)_i = \sum_j \frac{(\operatorname{mult}_F f^* E_j) a_j}{r_F} \mathbf{z}_j, \tag{6.7}$$

which is a convex combination of the $\mathbf{z}_j$ with $\operatorname{mult}_F f^* E_j > 0$.

Let S be the sum of the components F of Σ with

$$x \in g(F), \qquad \left(\frac{\operatorname{mult}_F D_i}{a_F} \right)_i = p^0, \qquad r_F = a_F.$$

Since $p^0 \ne 0$, F_* is a component of Σ. By **Step 1**, $r_{F_*} = c = a(F_*, X, 0)$, so F_* is a component of S and $S \ne 0$. Since S is a sum of components of Σ, **Step 3** implies that $\operatorname{Supp} \Sigma = \operatorname{Exc}(g) \cup \operatorname{Supp}(B + g^* F_0 + S + E_y)$.

Let F be a component of S. Since $r_F = a_F > 0$, (6.6) and (6.7) imply that $x \in h(W_F)$ and $p^0 \in C_{W_F}$. By **Step 1**, $h(W_F) = \{x\}$. Thus $g(F) = \{x\}$. Since $x \notin \operatorname{Supp} D'$, we have $\operatorname{mult}_F g^* D' = 0$. By (6.5), $\operatorname{mult}_F G = n a_F$. Since $b^0 \cdot p^0 = 1$,

$$\operatorname{mult}_F B = (1 - \eta) a_F (b^0 \cdot p^0) + \eta a_F = a_F.$$

This proves (2).

Let F be a component of Σ which is not a component of S, with $F \cap S \ne \emptyset$. Let F' be a component of S with $F \cap F' \ne \emptyset$ and $z \in F \cap F'$ a closed point. Since $g(F') = \{x\}$ and $g(\operatorname{Supp} S) \cap \operatorname{Supp} D' = \emptyset$, we have $g(z) = x$ and $\operatorname{mult}_F g^* D' = 0$. Since $r_{F'} > 0$, $f(z)$ belongs to some E_j. Let W be the stratum of $\sum_j E_j$ which is the irreducible component containing $f(z)$ of the intersection of the E_j containing $f(z)$. Then $x \in h(W)$. If $\operatorname{mult}_F f^* E_j > 0$ or $\operatorname{mult}_{F'} f^* E_j > 0$, then $f(z) \in E_j$, so $W \subset E_j$. Thus (6.7) for F' implies that $p^0 \in C_W$.

If $r_F = 0$, then $\operatorname{mult}_F f^* E_j = 0$ for all j, so $\operatorname{mult}_F D_i = 0$ for any i and $\operatorname{mult}_F G = 0$. Thus $\operatorname{mult}_F B = 0 < a_F$.

Assume that $r_F > 0$, and let $p^F := (\operatorname{mult}_F D_i / r_F)_i$. By (6.7), $p^F \in C_W$. By (ii) in **Step 1**, $b^0 \cdot p^F \le 1$. Let

$$\theta := \min_{i \in \mathcal{I}} \frac{p_i^F}{p_i^0}.$$

By (6.5), $\operatorname{mult}_F G = n r_F \theta$. Since $\operatorname{mult}_F g^* D' = 0$,

$$\operatorname{mult}_F B = r_F \left((1 - \eta)(b^0 \cdot p^F) + \eta \theta \right).$$

Since $\sum_{i \in \mathcal{I}} b_i^0 p_i^0 = b^0 \cdot p^0 = 1$, $b_i^0 > 0$ for any i, and $p_i^F \ge 0$ for any i, we have

$$\theta = \sum_{i \in \mathcal{I}} b_i^0 p_i^0 \theta \le \sum_{i \in \mathcal{I}} b_i^0 p_i^F \le b^0 \cdot p^F \le 1. \tag{6.8}$$

Thus $\operatorname{mult}_F B \le r_F \le a_F$. Suppose that $\operatorname{mult}_F B = a_F$. Then $r_F = a_F$, and all the inequalities in (6.8) are equalities with $\theta = 1$. Since $b_i^0 > 0$ and $p_i^F \ge p_i^0$ for $i \in \mathcal{I}$, the first inequality implies that $p_i^F = p_i^0$ for $i \in \mathcal{I}$. Since $b_i^0 > 0$ for any i, the second inequality implies that $p_i^F = 0 = p_i^0$

for $i \notin \mathcal{I}$. Thus $p^F = p^0$. Since $x = g(z) \in g(F)$, F is a component of S, a contradiction. Thus $\operatorname{mult}_F B < a_F$. This proves (3). □

6.3. **Lifting sections.** In this subsection, we prove Proposition 6.5 and Proposition 6.1.

Proposition 6.5 (cf. [Nug26, Proposition 6.3])**.** *Notation and conditions as in Set-up 6.2. Let $q > t$ be a rational number, M a Cartier divisor on X, and $F_0 \geq 0$ a $\mathbb{Q}$-divisor on X with $M - qL \sim_{\mathbb{Q}} F_0$ and $x \notin \operatorname{Supp} F_0$. Then there exists $\sigma \in H^0(X, K_X + M)$ with $\sigma(x) \neq 0$ and $\sigma(y) = 0$.*

Proof. By Lemma 6.4, there exist $g : Y' \to X$, B, S, E_y, and Σ such that Lemma 6.4(1)–(4) hold for q and F_0. Let $B_0 := B + g^*F_0$. Then $B_0 \geq 0$ has SNC support, and

$$g^*M - B_0 \sim_{\mathbb{Q}} qg^*L - B$$

is nef and big by Lemma 6.4(1). Let

$$P := K_{Y'} + g^*M - \lfloor B_0 \rfloor.$$

Since $P - K_{Y'} \equiv (g^*M - B_0) + (B_0 - \lfloor B_0 \rfloor)$ and $(Y', B_0 - \lfloor B_0 \rfloor)$ is klt ($B_0 - \lfloor B_0 \rfloor$ has SNC support and coefficients in $[0, 1)$, [KM98, Corollary 2.31(3)]), [KM98, Theorem 2.70] and Serre duality imply that $H^1(Y', \mathcal{O}_{Y'}(P)) = 0$. Thus the restriction map

$$H^0(Y', \mathcal{O}_{Y'}(P + S)) \to H^0(S, \mathcal{O}_S(P + S))$$

is surjective.

Let

$$C_0 := K_{Y'/X} - \lfloor B_0 \rfloor + S,$$

so that $P + S = g^*(K_X + M) + C_0$. Recall that $\operatorname{mult}_F K_{Y'/X} = a(F, X, 0) - 1$ and that $a(F, X, 0)$ is an integer for any prime divisor F on Y'. Let F be a component of S. By Lemma 6.4(2) and $x \notin \operatorname{Supp} F_0$, we have $\operatorname{mult}_F g^*F_0 = 0$, so $\operatorname{mult}_F B_0 = a(F, X, 0)$ and

$$\operatorname{mult}_F C_0 = a(F, X, 0) - 1 - a(F, X, 0) + 1 = 0.$$

Let F be a component of Σ which is not a component of S, with $F \cap S \neq \emptyset$. Since $g(F)$ contains x, we have $\operatorname{mult}_F g^*F_0 = 0$, so by Lemma 6.4(3), $\operatorname{mult}_F B_0 < a(F, X, 0)$. Thus $\lfloor \operatorname{mult}_F B_0 \rfloor \leq a(F, X, 0) - 1$ and $\operatorname{mult}_F C_0 \geq 0$. Since C_0 is supported on Σ, $C_0 \geq 0$ in a neighborhood of S, and no component of S is a component of $\operatorname{Supp} C_0$.

Since S is reduced and $g(S) = \{x\}$, $g|_S$ factors through the reduced point x. Thus a non-zero element of $(K_X + M)|_x$ pulls back to a nowhere vanishing section of $\mathcal{O}_S(g^*(K_X + M))$. Multiplying it by the canonical section of $\mathcal{O}_{Y'}(C_0)$ near S, we obtain a section of $\mathcal{O}_S(P + S)$ which does not vanish at the generic point of any component of S. Let $\tau \in H^0(Y', \mathcal{O}_{Y'}(g^*(K_X + M) + C_0))$ be a lift of that section.

The section τ satisfies the conditions of [OAI26, Lemma 2.5]. For any component F of S, we have $g(F) = \{x\}$, $\operatorname{mult}_F C_0 = 0$, and τ does not vanish at the generic point of F. For a prime divisor F on Y' which is not exceptional over X and is not a component of S, we have $\operatorname{mult}_F K_{Y'/X} = 0$, and since $B_0 \geq 0$, $\operatorname{mult}_F C_0 = -\lfloor \operatorname{mult}_F B_0 \rfloor \leq 0$. Thus the positive part of C_0 is exceptional over X, and by [OAI26, Lemma 2.5], τ is the pullback of a section $\sigma \in H^0(X, K_X + M)$ with $\sigma(x) \neq 0$.

By Lemma 6.4(2) and (4), E_y is not a component of S. Since $B_0 \geq B$, by Lemma 6.4(4), $\operatorname{mult}_{E_y} B_0 > a(E_y, X, 0)$. Thus

$$\operatorname{mult}_{E_y} C_0 \leq a(E_y, X, 0) - 1 - a(E_y, X, 0) = -1.$$

Since τ is a section of $\mathcal{O}_{Y'}(g^*(K_X + M) + C_0)$,

$$\operatorname{mult}_{E_y} \operatorname{div}(g^*\sigma) \geq -\operatorname{mult}_{E_y} C_0 \geq 1.$$

Thus σ vanishes on $g(E_y)$. By Lemma 6.4(4), $\sigma(y) = 0$. □

In Proposition 6.5, the point y may belong to the support of F_0, because the pullback of F_0 only increases the coefficient along E_y, so in the proof of Proposition 6.1 below, we remove $\operatorname{Supp} F_0$ only from the choices of x.

Proof of Proposition 6.1. Since L is big, U is a non-empty open subset of X [ELM+06, Example 1.7]. By Lemma 2.4(3), there exists a positive integer k such that for any integer $m \geq k$, $|mL|$ has no base point in U and separates any two distinct closed points of U. Let

$$U^\circ := \bigcap_{m \geq k} U_m^\circ \setminus \operatorname{Supp} F_0.$$

By Lemma 4.4(3), U° is the complement in $U \setminus \operatorname{Supp} F_0$ of countably many proper Zariski closed subsets of X. Since $\mathbb{C}$ is uncountable, U° intersects any non-empty open subset of X.

Let $x \in U^\circ$. If $q > d+1$, then we let $t := (d+1+q)/2$, so $d+1 < t < q$, and by Proposition 5.1(1), there exists an integer $m \geq k$ such that (m,t) is good at x. Otherwise, $L^d \geq 2$, and by Proposition 5.1(2), there exist a real number t with $0 < t < d+1 \leq q$ and an integer $m \geq k$ such that (m,t) is good at x. Since $x \in U_{m'}^\circ$ for all integers $m' \geq k$, we have $x \in U_m^\circ$. By Proposition 6.5, for any closed point $y \in U \setminus \{x\}$, there exists $\sigma \in H^0(X, K_X + M)$ with $\sigma(x) \neq 0$ and $\sigma(y) = 0$.

Let $U_0 := U \setminus \operatorname{Bs}|K_X + M|$, and let $\varphi : U_0 \to \mathbb{P}^N$ be the morphism defined by $|K_X + M|$. Since $d \geq 1$, $U \neq \{x\}$, so $x \notin \operatorname{Bs}|K_X + M|$. Since $x \in U^\circ$ is arbitrary, $U^\circ \subset U_0$. For any closed point $y \in U_0 \setminus \{x\}$, the section σ above defines a hyperplane of $\mathbb{P}^N$ which contains $\varphi(y)$ and does not contain $\varphi(x)$. Thus for any $x \in U^\circ$, x is the only point of $\varphi^{-1}(\varphi(x))$.

Suppose that φ is not a birational map onto the image, and let T be the closure of $\varphi(U_0)$. If $\dim T < d$, then by [Har77, Exercise II.3.22(b)], there exists a non-empty open subset T_0 of T such that any non-empty fiber of φ over T_0 has positive dimension. If $\dim T = d$, then by [Har77, Exercise II.3.7] and [Har77, Corollary III.10.7], there exists a non-empty connected smooth open subset T_0 of T such that φ is finite and étale over T_0. The number of points of the fibers of φ over T_0 is constant, and it is at least 2 because φ is not birational onto the image. In both cases, any point of the non-empty open subset $U_1 := \varphi^{-1}(T_0)$ of U_0 has another point of U_0 in the fiber of φ through it. Since $U^\circ \cap U_1 \neq \emptyset$, this is a contradiction. □

7. Proofs of the main theorems

In this section, we prove Theorems 1.5 and 1.6.

Since K_X may fail to be $\mathbb{Q}$-Cartier, we do not pull back K_X, and we push sections forward instead. The sections obtained on X form a linear subsystem. A linear subsystem is enough for birationality, because enlarging a linear system preserves birationality onto the image, but it yields no information on base points or on separation of points.

Lemma 7.1 (cf. [Han26, Proposition 5.6])**.** *Let X be a normal projective variety, $h : X' \to X$ a resolution, and M' a Cartier divisor on X'. Assume that $|K_{X'} + M'|$ defines a birational map onto the image. Then $|K_X + h_*M'|$ defines a birational map onto the image.*

Proof. Since X is normal, h is an isomorphism over an open subset of X whose complement has codimension at least 2. Thus $h_*K_{X'} = K_X$. A non-zero element of $H^0(X', K_{X'} + M')$ is a rational function f such that $\operatorname{div}(f) + K_{X'} + M' \geq 0$. Pushing forward by h, we obtain

$$\operatorname{div}(f) + K_X + h_*M' \geq 0.$$

The inequality is only a condition in codimension one on X. Since $\mathcal{O}_X(K_X + h_*M')$ is a reflexive sheaf, it is determined by the restriction to the complement of any closed subset of codimension at least 2 [Har80, Proposition 1.6]. Thus f is a global section of $\mathcal{O}_X(K_X + h_*M')$, and

$$H^0(X', K_{X'} + M') \subset H^0(X, \mathcal{O}_X(K_X + h_*M')).$$

Take a basis $s_0, \dots, s_{N'}$ of $H^0(X', K_{X'} + M')$ and extend it to a basis $s_0, \dots, s_N$ of $H^0(X, \mathcal{O}_X(K_X + h_*M'))$. Let $g : X \dashrightarrow \mathbb{P}^N$ and $g' : X' \dashrightarrow \mathbb{P}^{N'}$ be the maps defined by

$s_0, \dots, s_N$ and $s_0, \dots, s_{N'}$, and let $p : \mathbb{P}^N \dashrightarrow \mathbb{P}^{N'}$ be the linear projection that forgets the extra coordinates. Then $g' = p \circ g \circ h$. Since g' and h are birational onto the image, g is birational onto the image. □

Proof of Theorem 1.5. Let $h : X' \to X$ be a projective resolution and let $L' := h^*L$. Since $D - (d+1)L$ is big and the big cone is open [Laz04, Theorem 2.2.26], there exists a rational number $q > d+1$ such that $D - qL$ is big. Thus $h^*D - qL'$ is big [Laz04, Proposition 2.2.43], and there exists a $\mathbb{Q}$-divisor $F_1 \geq 0$ on X' such that $F_1 \sim_{\mathbb{Q}} h^*D - qL'$.

Let $M := \lceil h^*D \rceil$ and $R := M - h^*D \geq 0$ (cf. [Han26, Proposition 5.6]). Then

$$M - qL' \sim_{\mathbb{Q}} F_1 + R.$$

Proposition 6.1 for (X', L') with $F_0 = F_1 + R$ implies that $|K_{X'} + M|$ defines a birational map onto the image. Since X is normal, h is an isomorphism over an open subset of X whose complement has codimension at least 2. Thus $h_*\lceil h^*D \rceil = \lceil D \rceil$, and Lemma 7.1 with $M' = M$ implies that $|K_X + \lceil D \rceil|$ defines a birational map onto the image. □

Proof of Theorem 1.6. (1) Let $m \geq d+3$ be an integer, $L := K_X$, and $D := (m-1)K_X$. Then $D - (d+1)L = (m-d-2)K_X$ is big, and Theorem 1.5 implies that $|K_X + \lceil D \rceil| = |mK_X|$ defines a birational map onto the image.

(2) Let $m \geq d+2$ be an integer. Let $h : X' \to X$ be a projective resolution and let $L' := h^*K_X$. Then L' is nef and big, and $L'^d = K_X^d \geq 2$. Since $m - 1 \geq d+1$, Proposition 6.1 for (X', L') with $q = m-1$, $M = (m-1)L'$, and $F_0 = 0$ implies that $|K_{X'} + M|$ defines a birational map onto the image. Since $h_*M = (m-1)K_X$, Lemma 7.1 with $M' = M$ implies that

$$|K_X + (m-1)K_X| = |mK_X|$$

defines a birational map onto the image. □

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